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Project title	PotATo crop effective mAnagement strategies to tackle Future pEST threats
Acronym	**PataFEST**
Topic	HORIZON-CL6-2022-FARM2FORK-02-02-two-stage: Emerging and future risks to plant health
Type of action	Research and Innovation Action (RIA)
Duration	48 months

List of participants

Nº	Participant Organisation Name	Key Role	Type	Country
1	Fundación para el Desarrollo y la Innovación Tecnológica [**FUND**]	Coordinator; postharvest biocontrol solutions AI algorithm prediction model; portable volatiles sensor	RTO	ES
2	Instituto Vasco de Investigación y Desarrollo Agrario [**NEIK**]	Resistance genes identification; molecular characterisation	RTO	ES
3	Instituto Nacional de Investigaciones Agropecuarias [**INIAP**]	Pest pathway monitoring; characterisation of germplasm; field and bioassays	RTO	EC
4	IRIS Technology Solutions LS [**IRIS**]	Modelling and data processing. Image analysis digital tool	SME	ES
5	Kneia SL **[KNEIA]**	Diss, Comm., Exploit & Citizen Science	SME	ES
6	Cranfield University [**CRUN**]	Volatile biomarkers identification. Storage technology; plant soil management	UNI	UK
7	Udapa S. Coop. [**UDAPA**]	Validator of plant soil management strategies	SME	ES
8	Chemia Spa [**CHEM**]	Plant and postharvest solutions manufacturer	SME	IT
9	Palacký University Olomuc [**UPOL**]	Pest pathway monitoring and microbial ecology; Phenotyping study for portable pest early detection tools	UNI	CZ
10	Technion-Israel Institute of Technology [**TECHN**]	Vector repellent and antifungal solution	UNI	IL
11	European Potato Trade Association [**EUPAT**]	Multi-actor approach. Diss&Comm.	ASS	BE
12	Alma Mater Studiorum Universitá di Bologna [**UNIBO**]	Biobased solutions for soil management	UNI	IT
13	Kartoffel-Boehmer [**BOEH**]	Validator of plant soil management and postharvest strategies	SME	DE
14	European Citizen Science Association [**ECSA**]	Citizen Science approach	ASS	DE
15	Anecoop [**ANEC**]	Multi-actor approach. Diss&Comm.	LE	ES
16	Terrea [**TERREA**]	Validator of plant soil management and postharvest strategies	SME	FR
17	Albert Bartlett [**AB**]	Validator of plant soil management and postharvest strategies	LE	UK
18	Ainia [**AINIA**]	LCA, LCC, s-LCA assessment	RTO	ES

Acronyms: BC: *Bactericera cockerelli;* **CA**: controlled atmosphere*;* **CLso**: *Candidatus* Liberibacter solanacearum*;* **IPM**: Integrated pest management; **IR**: Infrared; **LC-MS**: liquid chromatography-mass spectrometry; **ha**: hectares **PHTS**: Phenotype-based High-throughput screening; **RGs**: resistant genes; **SNPs**: single nucleotide polymorphisms; **tons**: tonnes; **VOCs**: volatile organic compounds; **ZC**: Zebra chips.

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1 EXCELLENCE

The **potato (*Solanum tuberosum*)** is a key crop in Europe. In 2020, the **EU's potato production was 55.3 million tonnes** (worth 12.3 billion euros), with Germany (21.2%), Poland (16.4%), France (15.7%), and the Netherlands (12.7%) being the top EU producers. Spain had a production of 2.1 million tonnes (2020) and UK, according to FAO, was ranked in the 14th position of top countries in potato production (5.2 million tonnes, in 2019).[1] In South America, cultivated potatoes are widely distributed in the Andean region with over 4,500 varieties and more than 100 wild species. In **Ecuador**, **total potato production in 2017 was 400,000 tonnes** where the cultivation is generally undertaken by small-scale farmers[2].

Potato is **vulnerable to a variety of pests and soil-borne pathogens**, resulting in significant economic losses in the potato industry, and consequently food losses (in line with SDG 12.3)[3,4]. For example, in 2011, the United States recorded **losses in potato yield of up to 75%,** and in other cases, the damages are so severe that the entire crop is lost[5]. In view of this scenario, the European regulatory framework for potato's imports and production sets strict requirements. For example, the European and Mediterranean Protection Organization (EPPO) establishes phytosanitary measures which should be required for seed and ware potatoes moving in international trade to prevent the introduction and spread of quarantine pests[6]. Although there are strict controls, globalisation, climate change and economical drivers are threatening the entrance of emerging or new potato pests from imported potato to EU. The most important **pest in the potato plant** is caused by **bacteria *Candidatus* Liberibacter solanacearum (CLso)** and which disease is known as **zebra chip (ZC)**. The introduction the CLso bacterium on potato is mainly linked to the presence of the **vector psyllid *Bactericera cockerelli* (BC)[7].** Additionally, other threats (such as ***Fusarium sambucinum, Colletothricum coccodes, Helminthosporium solani***) also affect potato safety and quality **during the postharvest activities**, making the tubers unmarketable[8]. All the above evidence establishes the need to start the project PataFEST as currently there is not EU collaborative project tackling these potato diseases.[9]

To preserve the health of potato plant against its emerging pests and the quality of the final product, the project **PataFEST** will (Fig 1.1): **1) characterize at molecular level** the ecological pest spread pathway and identify potato disease resistance varieties against CLso and postharvest pathogens; **2)** provide effective **preharvest plant and soil treatments** against CLso vector and soil-borne pathogens combined with other cutting-edge digital technologies such as **image analysis tools (mobile app)** and **artificial intelligence predictive models**; **3)** develop **postharvest technologies** (**biocontrol coating solution**, **controlled atmosphere storage** and volatile organic compounds (**VOCs) sensors**) to control the incidence of soil pathogens and maintain the quality of potato tubers stored. All these three strategies and technologies will be incorporated into an **integrated pest management (IPM)** to validate their performance in a country-specific relevant environments (TRL5) in **Germany**, **France**, **Spain** and **UK** as representatives of main potato producers. In addition, **Ecuador** will host field trials against CLso and its vector BC. PataFEST will ensure a long-term strategy for tackling future potato pest threats by carrying out strong dissemination, communication and exploitation plan, with an overarching **multi-actor approach** and **citizen of science** activities.

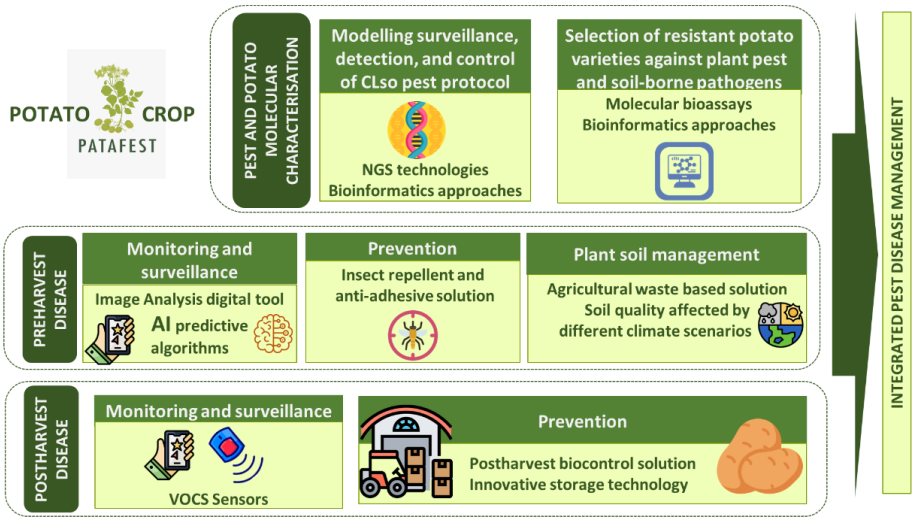

Figure 1.1 PataFEST at a glance

[1] Top Countries in Potatoes Production (2019). Available here.
[2] Rondon, S.I. (2022). In: Alyokhin, A. et al. ed. Insect Pests and Potato. 317-330.
[3] The EU potato sector – statistics on production, prices, and trade. July 2021. Available here.
[4] SDG 12.3: By 2030, halve per capita global food waste at the retail and consumer levels and reduce food losses along production and supply chains, including post-harvest losses.
[5] Greenway, G. (2014). Am. J. Potato Res. 91: 714–719.
[6] PM 8/1 (2) Potato. (2017). Bulletin OEPP/EPPO, 47(3): 487–503.
[7] Mora, V., et al. (2021). Frontiers in Microbiol. 12: 700663.
[8] Wood, et al. (2013). Biol. Control. 64: 152-159.
[9] Check in Cordis during the proposal preparation.

1.1 Objectives and ambitions

1.1.1 Objectives

The goal of PataFEST is to reduce: 1) the risk of introducing and spreading emerging potato pests that could endanger EU agriculture and 2) the presence of soil-borne pathogens in the potato postharvest activities. The selected threats in this project are: 1) ***Candidatus* Liberibacter solanacearum** and its vectors (**BC**, Fig. 1.2) affecting potato and causing **ZC disease** (Fig 1.3) within the crop cycle (**preharvest threat**) and 2) soil-borne pathogens such as ***Fusarium sambucinum***, ***Colletothricum coccodes***, and ***Helminthosporium solani*** causing high incidence of rot and skin blemish on potato tubers during postharvest storage (**postharvest threat**).

Figure 1.2. Vector psyllid

Zebra Chip disease causes plant symptoms such as **leaf chlorosis**, **discoloration**, **curling** and eventually **premature plant death**. The potato chip industry is also affected by this disease since the CLso infected potato tubers are often deformed, of low quality and, with dark stripes, making **chips bitter to taste** and unacceptable for manufacturers to commercialise. CLso has been found throughout North and South America, Canada, New Zealand, and Australia and the disease is reported in other solanaceous crops such as tomato, eggplant, and pepper [7,10]. In **Ecuador**, BC was first reported in 2019[11] and CLso in 2020[12], as a result, **up to 100% potato yield losses** were reported whilst the **cultivated area was reduced by 56%** compared with the previous years[13]. Preventative measures are implemented to decrease the potential appearance of ZC vector in European potato supply chain. For example, EPPO classified ZC as an A1 quarantine pest (pest that is not found in the EPPO territory, but it is recommended to carry out a quarantine management strategy)[14]. However, considering the potential threat of this emerging pest, the development of further management strategies for prevention, early detection and surveillance is required.

Figure 1.3. Zebra Chip infected potato

Chemical pesticides are used to control pest and diseases to secure yields in plant production and reduce food wastage. The **EU Green Deal aims to reduce fossil-based pesticides by 50%** by implementing more sustainable and circular plant health solutions[15]. Possible consequences derived from the excessive use of chemical pesticides and the transition towards their bio-based counterparts are 1) increases **pesticide resistance** in bacteria, insects, and soil-borne pathogens, and 2) **reappearance of soil pathogens**, potentially increasing the incidence of soil infections during postharvest storage.

Main **potato postharvest diseases** include **dry rot caused by *Fusarium sambucinum*** (Fig. 1.4), black dot caused by ***Colletothricum coccodes***, and silver scurf provoked by ***Helminthosporium solani***. These pathogens are soil-borne and have incidence in both field and storage stages with dramatic economic losses for potato farmers and pre-pack potato industry since the postharvest development of such diseases will result in unmarketable product[16]. Standard control for these diseases in warehouses includes elimination of infected tubers prior to storage and storage management strategies such as forced air ventilation, and controlled temperature and humidity feedback systems[17]. Chlorine-based disinfectants are also applied. However, chlorine may not completely prevent storage pathogens, and the presence of chlorinated residues (chlorates and perchlorates) may cause consumer worries, prompting them to demand more environmentally friendly and chemical-free alternatives, particularly for organic farming.

Figure 1.4. Dry-rot infected potato

The project PataFEST is aligned with the topic *Horizon-CL6-2022-FARM2FORK-02-02 two stages: Emerging and future risks to plant health* promoting the transition to a fair, healthy, and resilient agriculture. To achieve this, the proposed strategies, in lined with the topic's scope, are:

Molecular characterisation of the targets potato pests and diseases by studying the spread pathways and determining potato resistance genes by using innovative molecular biology methodologies (NGS sequencing), bioinformatics tools and biostatistical analyses.

Early detection and surveillance of pest and pathogen diseases by applying real-time diagnosis to potato plants and tuber health using high performance and user-friendly portable sensors and smartphone apps. Artificial intelligence (AI) predictive algorithms will be developed as a tool for the classification of the pest diseases integrated in a mobile app and forecasting potential risk context of disease spread caused by climate change and globalisation.

Prevention methodologies for potato plant, tubers and soil health by developing chemical free bio-based sustainable solution to hinder the spread of pests and diseases.

[10] https://gd.eppo.int/taxon/PARZCO/distribution. Last updated 2021-10-05.
[11] Castillo, C.I., et al. (2019) Bull. Insectol. 72: 85-91.
[12] Caicedo, J.D., et al. (2020) Australasian Plant Dis. Notes 15, 6.
[13] MAG-SIPA http://sipa.agricultura.gob.ec/index.php/papa.
[14] PM 1/1 (30) EPPO A1 and A2. (2021). Lists of pests recommended for regulation as quarantine pests.
[15] European Green Deal. Available here.
[16] Powelson, M.L., Rowe, R.C. (2008). In: Johnson D., ed. Potato Health Management. 183-195.
[17] Knowles N.R., Plissey E.S. (2008). In: Johnson D., ed. Potato Health Management. 79–99.

Integrated pest management strategy by validating potato disease gene resistance varieties, surveillance and early detection tools, and environmentally sound treatments **in plant, tubers and soil** of both organic and conventional crops. The validations will be done under relevant environment (**TRL5**), will be at European and international (Ecuador) scale (Table 1.1).

Table 1.1 Proposed validation trials in selected EU countries and Ecuador (TRL 5).

Pests and pathogens	Potato varieties	Trials size	Country	Partner
Plant pest and its vector *Candidatus* Liberibacter solanacearum (CLso) *Bactericera cockerelli* (BC)	30 potato varieties	2 ha of plant potato field	Ecuador	INIAP
Soil-borne pathogen: *Fusarium sambucinum,* *Colletothricum coccodes* *Helminthosporium solani*	Common potato varieties: Vivaldi, La Vie. Other potato varieties: Monalisa (Spain); Maris Piper (UK); Gala/Wega (Germany); Melody (France).	1 ha of plant and soil potato crop; 1 ton potato storage	Spain UK Germany France	UDAPA AB BOHE (organic) TERREA (organic)

PataFEST project will boost a **multi-actor approach and transnational research** collaborations among **18 partners** with high expertise in molecular biology, potato plant protection, postharvest technologies, engineering for agriculture applications, soil health and pest and disease management. In addition, **citizen science** will be used as a tool to monitor emerging potato threats and collect scientific data by the general public.

The project proposes **5 objectives (O)** with their respective **5 outcomes (OC)**, articulated under **18 results (R).**

O1. To develop molecular approaches for the pest surveillance protocol and disease-RGs identification in potato varieties. Including: 1/ to develop of molecular monitoring protocol for the surveillance, detection, and control of CLso plant pest; 2/ to setup a comprehensive catalogue of mapped RGs against CLso pest disease and soil-borne postharvest pathogens; 3/ to validate the RGs in the working collection of potato accessions by means of field trials, bio-assays and association mapping techniques using the detected SNP.

R1	150-200 high quality bacterial **genomes**, 500-1000 **metagenomics** 16S samples and 1500-2000 **qPCR assays** will be performed to carry out the **whole monitoring** and **control protocol** of **CLso spreading** and its **vector**.
R2	Full **identification**, **compilation**, and **mapping** against the reference genome of all **annotated potato RGs**, QTLs and orthologous genes from the Plant Resistance Genes database (PRGdb).
R3	A broad **germplasm working collection** of around 500 samples will be **analysed**. 40 wild species, 35 native varieties and around 280 old and new **European cultivars** will be characterised for postharvest disease resistance levels. In addition, 150 **samples from Ecuador** (advanced breeding materials, wild species, native varieties, European cultivars) which will be **characterised** for their **tolerance level to** both, **CLso** and its **vector**.
R4	**Identification** of **promising RGs** SNP for **targeted diseases and pests** which can be applied in marker-assisted selection within molecular potato breeding. Determination of breeding values for pest and disease resistance of the accessions.
OC1	PataFEST project will provide relevant and free access information to better identify the most suitable potato varieties resistant to plant pest threat and soil-borne pathogens in postharvest storage.
O1's pertinence to the call requirements: *"Increase knowledge of the biology, pathways of entry, behaviour in the plant-soil system..."; "Identify and introduce resistance traits to support the long-term sustainability of agriculture and forestry in the EU and/or Associated Countries".*	

O2. To develop innovative strategies to control potato pest spread and reduce potato postharvest disease. Including: 1/ to develop low-cost diagnostic solutions as VOCs sensor devices and digital tool application (e.g. smartphone) for the early detection and surveillance of the plant pest disease and postharvest pathogens in potato warehouses; 2/ to detect and classify diseased potatoes and predict risks of disease spread due to climate change using AI.; 3/to provide natural and sustainable solutions to prevent pest spread by controlling the vector in potato plants and investigating biocontrol and controlled atmosphere strategies to reduce postharvest disease incidence and severity; 4/ to reduce the incidence of CLso and post-harvest pathogens in potato crop increasing plant health status thanks to optimised nutrient availability and improved plant soil management.

R5	Morphological and physiological **phenotyping traits** (biomarkers) will be identified in infected plants and tubers through **Phenotype-based High-throughput Screening** (PHTS) which uses different non-invasive sensors as RGB, FluorCam, IR, and hyperspectral cameras. Besides, VOCs biomarkers will be identified and quantified by chromatography techniques in infected tubers. The best identified biomarkers will be selected

	and models for the **development** of "low cost", user-friendly and robust **digital tool** (smartphone app) and **portable sensor devices (VOCs sensor)** will be developed.
R6	A suite of **algorithms** and techniques based on **artificial intelligence** will be developed for the detection of pest and diseases in potato plants and potatoes using an **app mobile device**. Same technology will be also used to **predict** the potential **risks** context of **pest spread** due to **climate change** and **globalisation**.
R7	Environmentally friendly **wax sprays** will render potato leaves super hydrophobicity providing **anti-adhesive** performance against CLso bacteria vector (psyllid) and **anti-pathogenic properties**.
R8	Enzyme and antagonistic yeasts-based **biocontrol coatings** will be tested on potato tubers stored under different graduated CO_2 and O_2 controlled atmosphere (CA) conditions to **decrease** at least **40-50% the incidence of** some important **postharvest pathogen diseases** selected such as dry rot (*F. sambucinum),* black dot and silver scurf (*C. coccodes,* and *H. solani, respectively*) to ensure consumer safety and acceptance.
R9	Chitin and calcium carbonate-based **alternative biofertilizers** obtained from the valorisation of aquaculture biomass residues will be formulated as carriers of nutrient elements reducing susceptibility of potato plant to pathogens incidence both at foliar and tuber level. Soil management task will also include field trials to **evaluate pest pathway** of entry and **drivers of spread** as **future climate scenarios** (flooding, coldness, drought, etc.) and ecological degradation processes (modification of soil organic matter, etc.). **Soil management strategies will reduce pathogens incidence by 30%.**
OC2	Use of early detection tools, sustainable solutions to improve plant and soil health and control postharvest pathogens will be helpful for meaningful reduction of pesticide and fungicide usage in potato crops.
O2's pertinence to the call requirements: *"Develop and uptake rapid and effective tools for the prevention of entry, spread and establishment, early detection, surveillance, treatment and (bio) control of plant pests; "influence of climate change and globalisation of pest(s)".*	

O3. To validate the Integrated Pest Management. Including:1/ to implement the technologies developed and carry out plant bioassays trials against CLso pest disease in an intended environments beyond Europe (e.g., Ecuador); 2/ to validate the postharvest technology and soil treatment solution for both organic and convention potato crops at TRL5.

R10	At least **30 potato varieties** identified and selected for their resistant traits against CLso and its vector will be tested in field bioassay trials under real condition (potato plants affected by the bacteria in Ecuadorian geographical areas). Digital tool developed will be operated from a mobile hardware unit (smartphone or tablet) and validated in field performing **90% reliability** on the diagnostic detection of CLso disease in potato crops.
R11	Pesticide free natural solution performance will be formulated at pilot plant level and evaluated on potato plants (**2 ha**) to study the repellent effectiveness against BC in Ecuadorian geographical areas. The **presence of BC** will be **reduced by at least 50**% and the **incidence of ZC by 25%.**
R12	Biocontrol natural solution prototype will be formulated at pre-industrial level and tested on **2 tons** of tubers (organic and conventional) stored to control postharvest pathogen disease as F. *sambucinum, C. coccodes,* and *H. solani*. VOCs sensors developed will be tested in 3 different European potato warehouses under real storage conditions (at least 100Kg x warehouse). Trails will be carried out in warehouses located in France, Germany, and UK.
R13	Soil treatment solution based on aquaculture by-product biomass residues will be evaluated in **1 ha** of potato crop fields (organic and conventional) under real cultivation conditions in Spain, France, and Germany.
OC3	Carry out trials both in potato plant and in tubers during postharvest storage of the technologies developed at TRL5 will validate the effectiveness of the integration pest management to control emerging threats and disease for potato crops.
O3's pertinence to the call requirements: *"Sustainable and integrated pest management"; "International cooperation"; "Economic implication for farmer and ecological impacts of plant pests".*	

O4. To engage citizen as a source to collect scientific and technical data. Including: 1/ to organize citizen science campaigns and workshop to collect information database related to potato pest diseases; 2/ to organize webinars to increase the engagement of citizen in the project activities.

R14	Involving social groups (citizens as students, urban garden owner, scientists, potato production farmers) in such events will provide a low-cost means of maximizing research efforts across EU and abroad regions.
R15	**Three** types of **citizen science activities**: 1) 3 co-creative workshops to experience the sustainable solutions developed in the project by citizens with urban gardens and farmers; 2) 2 citizens campaign based on practical demonstration of molecular biology tools; 3) organise 8 webinars to disseminate the social implication of farmers and citizens.
OC4	The use of citizen science in the agriculture sector will be a fruitful opportunity to provide an effective supplement to pest management approaches.

O4's pertinence to the call requirements: *"Integrate citizen science as a tool to monitor emerging pests".*

O5. To disseminate, communicate and exploit the main achievements of PataFEST project: Including: 1/ to involve the scientific & technical community, companies, and citizens/consumers across the potato value chain; 2/ to create a multi-actor participatory platform ensuring the effective dissemination of project results and outcomes to appropriate target audiences; 3/to develop a set of clustering activities with existing projects and relevant European initiatives and platforms.

R16	Creation of **multi-actor participatory platform** (MAPP) to share knowledge, expertise, and background across the interdisciplinary consortium to generate at least **60 "practice abstracts"** to contribute to **EIP-AGRI repository**.
R17	Project **dissemination** results will be **shared** through **relevant channels** to support the **impact** and **exploitation outcomes** including websites, social media, training materials, social events, etc.
R18	**2 workshops** together **with similar EU projects** and **clusters** to increase the project's visibility and reach.
OC5	Providing the foundations and drivers for future development and implementation of IPM strategies by farmers and other key players in the event that these emerging potato pests outbreak in EU.
O5's pertinence to the call requirements: *"Proposals must implement the "multi-actor approach" including a range of actors to ensure that knowledge and needs from various sectors such as research, plant health services and the farming/forestry sector are brought together".*	

1.1.2 Ambition and progress beyond the state of art

1.1.2.1 Ambition and innovation potential

PataFEST project aims to provide a powerful European initiative for a sustainable IPM strategies to treat and control the presence of CLso pest and its vector in plant and the incidence of several soil-borne pathogens in the potato postharvest storage. Beyond the state of art, several innovative strategies will be implemented within the project:

Pest spread pathway and ecological analysis: The combination of different molecular novel approaches of high-throughput DNA Sequencing (HTS) technologies as whole-genome sequencing (WGS) and 16S metabarcoding techniques will provide a deeply knowledge of CLso and its vector (BC) spread pathway allowing to create a surveillance pest disease protocol. Furthermore, these findings will allow to optimize the sensitivity and accuracy of quantitative PCR (qPCR) assays ensuring the detection of novel, diverse, and uncommon CLso strains and consequently significantly improve the plant pest routine diagnostics.

Potato disease-RGs analysis: All the molecular information available of disease-RGs will be combined under a singular complete collection database. For the first time, an innovative experimental association mapping regarding the ZC disease will be carried out using novel targeted sequencing methods (SPET or Capture) and hybridisation technology with potato microchips. Furthermore, the generated combined information obtained in PataFEST will represent a universal and powerful tool for analysing any new emerging pest or disease in potato in the future, by simply applying additional phenotypic evaluations of resistance levels, since the molecular data will be the same.

Early detection tools: PHTS technology will be adapted to identify potato phenotype traits according to changes of selected biomarkers provoked by stress-inducing factors in plant models. By this way, it will be possible to diagnose with more accuracy and in real time the potato plant health and select the most suitable methodology for the developing of a prototype tool (mobile app) able to perform an early detection of a pest or a pathogen.

Postharvest volatiles organic compounds sensors: A lab study of VOCs emissions during the infection of the potatoes will reveal the nature of biomarkers and their concentration values as function of the potato state. With this information, PataFEST project will develop a portable prototype of array of sensors able to monitor the concentration of these VOCs in real-time, allowing the potato packer to take in advance the right decision about when to open the chambers and release the product to the market.

AI using algorithm prediction model: The usage of AI predictive algorithms allows the process of observation to be proactive rather than reactive, without necessarily human intervention. Automatically and periodic revisions of potato health will permit to win anticipation time to fight efficiently against a pest or a pathogen spread in potato caused to climate change. For this purpose, AI predictive algorithm will fulfill two functions: classification of diseased cases and potential risk context of disease spread. These predictive assessments will consider global warming as a cornerstone, considering variables which depend on the geographical situation and climate conditions.

Pests control solutions: The development of sustainable and innovative pre and postharvest solutions based on environmental-friendly ingredients and biotechnological agents will be a bio-based alternative to the excessive fossil-based compounds use in potato crops. PataFEST solutions will mainly provide protection against pest and pathogens, however their application will contribute to reduce respiration rate and dehydration both in potato plant and tubers.

PHTS rapid method will be used to support the quantitative testing of pre and postharvest formulation prototypes effects on pest and pathogen incidence.

Plant soil management: The innovation of the solution proposed in PataFEST will consist in the value improving of biomass residues by-products such chitin and calcium carbonate not used directly as fertiliser but as a carrier able

to deliver and improve availability of essential elements to potato plants. Besides, in PataFEST testing different soil conditions will deliver understanding of how manage future climate scenarios on disease development to be tested at relevant scales, yet under replicable conditions.

Integrated pest management: PataFEST project will promote the opportunity to evaluate the synergy of the technological solutions within IPM strategies both for organic and conventional potato plant and postharvest tubers at TRL5. Field and postharvest trials will involve the participation of potato packers, cooperative, trader and farmer associations.

1.1.2.2 Positioning and Technological Readiness Level (TRL)

Innovation & Partners Involved	TRL Initial → final	Justify current and final TRL
Specific and sensitive molecular detection and mapping (UPOL, INIAP).	3→5	Molecular technologies such as qPCR and 16S metabarcode high-throughput sequencing have been demonstrated to significantly improve detection. In PataFEST, the implementation of a pest monitoring system and the improvement of the qPCR assays will allow triggering the timely deployment of management measures.
Potato germplasm collection (NEIK, INIAP).	2→5	The available published data will be converged in one complete collection data base and a singular validation will be performed using novel targeting sequencing methods (SPET or Capture).
Phenotype-based High-throughput Screening technology (UPOL, IRIS, CRUN).	2→5	PHTS as a non-invasive real-time analysis has been used in several plant models. PataFEST research activities will allow to correlate potato phenotype and stress-inducing factors (e.g., study elicitors of biotic stress) like those provoked by pests and pathogens selected. This will allow to develop robust digital tools for the pest and pathogen diseases early detection.
VOCs sensors (CRUN, FUND)	3→5	Electrochemical sensors technology has been successful developed to detect VOCs and prevent the spoilage of other agro-food products as meat and mushrooms. In PataFEST, the identification of specific potato health biomarkers (including volatiles) will allow developing high-performance sensors able to provide a fast and early- stage detection of emerging diseases.
Postharvest biocontrol solution (FUND)	3→5	Biocontrol solutions have been previously tested on surface of other fruits and vegetables against pathogens as *Colletothricum spp*. In PataFEST the solutions will be tested against soil-borne pathogens affecting potato crops.
Plant protective wax solution (TECHN)	3→5	Multi-functional pesticide-free treatment has been successfully tested on plant leaves controlling pathogens and bacteria disease. In PataFEST wax-based formulation will be tested as foliar spray solution to repel CLso vector, inhibit of soil-borne pathogen and control plant dehydration.
Plant soil sustainable solution (UNIBO)	3→5	Seashells biomass residues have been already tested for the formulation of new advanced materials for other industrial applications. In PataFEST innovative formulations will be tested to improve the soil quality and plant health in potato crops.

1.2 Methodology

1.2.1 Concept and overall methodology

PataFEST is a 48-month RIA action project based on the following activities against the presence of CLso pest threat in potato plants and pathogens incidences in postharvest tubers (Fig. 1.5).

Pest spread pathway and ecological analysis

The first goal of PataFEST is developing **molecular modeling approaches to surveillance, detection, and control of plant pests (WP1)** that will be decisive to aid in understanding the epidemiological spreading of CLso pest. Therefore, these findings will create an effective pest-disease surveillance protocol and an early and effective plant pest detection. In this way, identifying the precise **CLso** strain type represents a key step as the haplotypes (A and B) express different disease symptoms[18]. Initially, UPOL will summarize all available genetic (16S barcode) and genomic information of **BC** and **CLso** stored in the public databases as GenBank. Furthermore, to cover all possible sub-haplotypes, the whole genome of the **CLso** sampled across Ecuador will be sequenced using highly accurate long-read from PacBio's next-generation sequencing platform. This approach will comprehensively characterize different strains and enable the efficient variant detection of plant pest. Additionally, genome-wide association (GWAS) analysis among all target genomes will be used to identify the diversity and potential differences in sites or protein families connected with the pest virulence. This approach will allow to review widely used primers

[18]Harrison, G.F., et al., (2019). Nat. Ecol. Evol. 3(8): 1253–1264.

and modify them to be more strain-specific or target other parts of the genome (with higher mutation rate, i.e., prophage section) during pest screening. Likewise, **the reconstruction of the spread pathway** across the target regions and phylogeny of CLso strains will be inferred using the metabarcoding approach across all assembled 16S rRNA genes from the vegetation near the fields of BC sampled. Combining inferred relationships among samples with the location and the collection date will provide complex data that will be analysed with Bayesian inference-based software (BioGeoBEARS). The spread pest pattern will also be tested using continuous trait reconstruction (BEAST and SPREAD3). This approach uses Google Earth to visualize the spread pathway. Besides all this, metabarcoding data will be used to identify the syn-positivity of **BC** for **CLso** and phytoplasmas. Using all our findings from previous steps, cost-effective diagnostic solutions based on qPCR will be developed to identify the samples positivity quickly. Finally, all generated data will be submitted to Genbank and available for open science.

Potato disease-RGs analysis

Taking into account that one of the main goals of PataFEST project is to access the most suitable potato varieties resistant to plant pest threat and soil-borne pathogens, several molecular approaches will be used for the **identification** and **characterisation of the RGs** and **the detection of functional genetic markers (WP1).** These findings are considered a crucial way to recognize and characterize the disease-resistance mechanisms to improve the selection of resistant varieties of potato at early stages of breeding and enhance agricultural production. First, a singular and complete catalogue of mapped RGs against potato pest and diseases will be addressed by NEIK using different tools, starting with the compilation of all annotated and published RGs on the potato reference genome. This initial compilation will be completed by mapping recently published potato RGs to the Potato Reference Genome. Besides, this collection will be extended by the compilation of all relevant RGs from the PRGdb, searching the potato orthologues; mapping and validating these genes to the Reference Potato Genome. Furthermore, published Potato Resistance QTLs will be mapped to the reference.

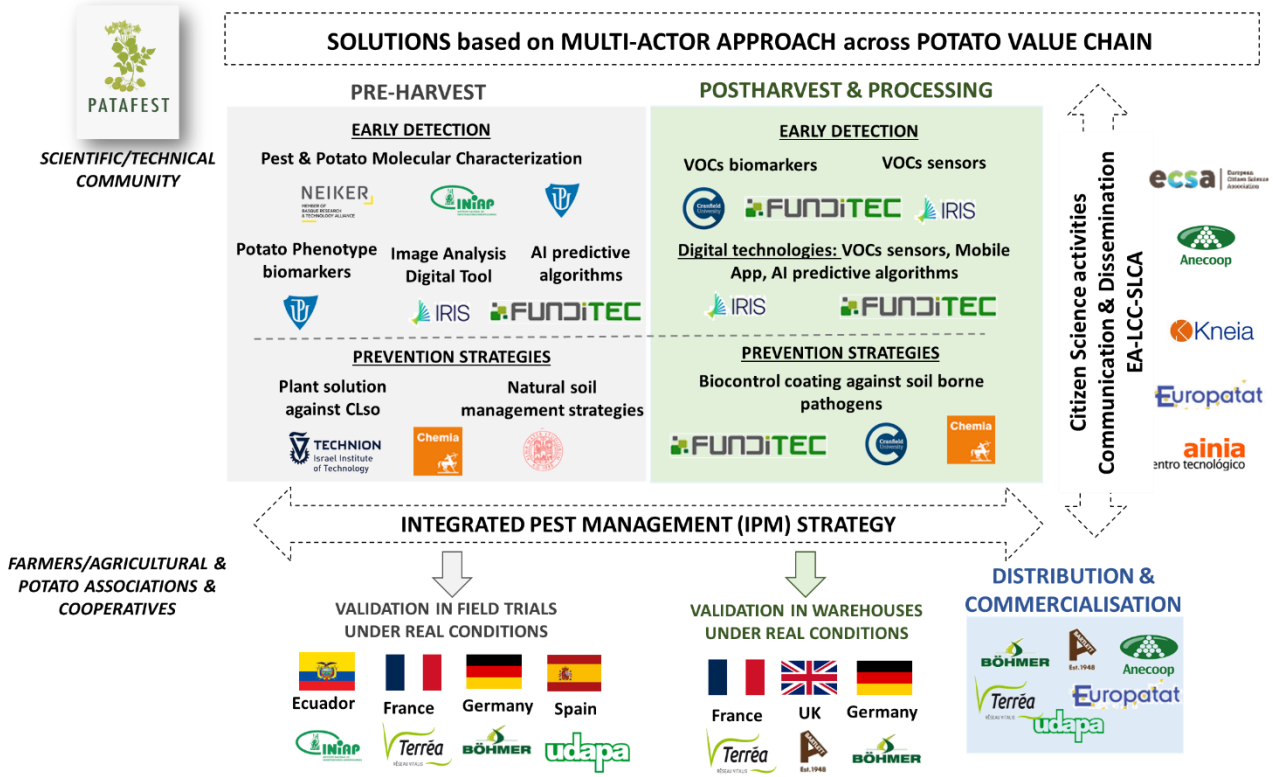

Figure 1.5. PataFEST general concept and participation of partners.

genome via projection, using collocated probes/primers with known sequences and identifying the potential collocated RGs that caused the QTL. Furthermore, one of the driving forces of PataFEST project is to do a **comprehensive molecular characterisation** approach of a broad germplasm collection with the view of havimg deeply information about their resistance and tolerance profiles. Initially, a molecular characterisation of the working collection of potato accessions will be achieved by NEIK using SNP GGP V4 Potato array, which contains already many RGs. Additionally, in order to enrich these preliminary results a relevant RGs will be analysed for SNP by using innovative and novel targeting sequencing strategies (SPET) with the Germplasm accessions. Moreover, field trails and bioassays will be performed with the aim to provide a global evaluation of resistance levels against CLso pest disease and soil-borne postharvest pathogens in the accessions. First the required in vitro plants and minitubers of local and introduced germplasm will be produce according to INIAP´s protocol using tissue culture or stem

cuttings, followed by hydroponic and aeroponics technologies [19,20]. Choice (antibiosis) and non-choice (antixenosis) experiments for BC will be established following a defined protocol[21,22]. Procedure and adapted methodologies will be performed to assess CLso resistance/tolerance[23]. Complementary field trails and bioassays of the accessions of the germplasm collection will be performed in order to characterize with respect to their resistance level to important diseases (and pests) that include dry rot caused by *F. sambucinum*, black dot caused by *C. coccodes*, and silver scurf provoked by *H. solani*. For all these soil-borne, defined susceptibility evaluation methodologies will be used [24, 25, 26]. Then, GWAS analysis between observed phenotypic resistance levels and molecular SNP data will be carried out for validating important RGs. The effects of each SNP on the resistance levels will be determined as well as the breeding values of each accession. These Association Studies will generate important and crucial information about the diversity existing in a germplasm collection, which allows for significant advances in the identification of accessions with potential for breeding programs. Finally, a **bioinformatic platform** for friendly data management will be deployed by NEIK to integrate all the previous results, and thus, to facilitate and encourage the **scientific divulgation** to the community **(WP1)**.

Early detection and surveillance tools

To develop **low-cost diagnostic solutions** based on an **image analysis application** for the early detection and surveillance of the potato pest disease and postharvest pathogens in the warehouses, **UPOL** will perform correlation of induced plant stress conditions with data obtained by a photometric system used in **PHTS approaches**[27,28] **(WP2).** The system based on RGB cameras, FluorCam, IR, and hyperspectral cameras will allow the study of the phenotyping changes of plant and tubers in non-invasive manner [29] (Fig. 1.6). The complex data obtained from the model situations will be used to monitor and correlate morphological and physiological changes with pest-induced plant stress conditions. **Multivariate statistical approaches** will be employed to **identify** non-invasive **morphological and physiological traits (biomarkers)** of individual diseases and the relevant sensor(s) to obtain them.

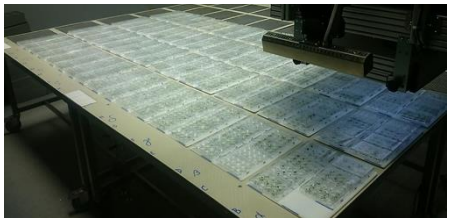

Figure 1.6. High-throughput phenotyping platform (UPOL facilities).

The **biomarkers** will be further **validated (WP2)** by **CRUN** in **glasshouse experiments** using the Plant Phenotyping and Soil Health Facility equipped by 1 m^3 lysimeter, in which soil physical conditions can be created to simulate realistic soil physical properties reflecting field conditions using a purpose build soil processor (Fig. 1.7). Each lysimeter is on balance to monitor water use, equipped with moisture and temperature sensors at 4 depths, and equipped with a fully automated irrigation system. Belowground cameras are used to monitor root and tuber development. Treatments include optimal water content, and suboptimal due to a prolonged period of drought during the growth period to assess the impact on growth, plant stress signals, belowground development and disease progression. Pathogens introduced will include *C. coccodes, H solani, and F. sambucinum*. **Plant growth** health will be **assessed** during growth at fortnightly intervals with **sensors** in the phenotyping platform. Sensors include **3D laser** (monitoring growth), **thermal** (stress, transpiration), **hyperspectral imaging** (pest and disease) and **RGB** (development, pest, and disease). Upon harvest, potato tubers will be evaluated using the same platform, which can achieve very detailed scans of hundreds of potatoes within minutes. Tubers will be moved into postharvest storage facilities and at set intervals evaluated using the same scanner to determine evolution of disease signals over time and disease severity upon harvest and development during storage will be correlated with imposed stress conditions during growth.

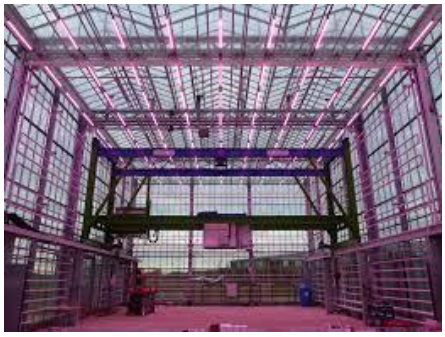

Figure 1.7. Plant Phenotyping and Soil Health Facility Soil Health Facility (CRUN facilities).

The **huge series** of the microtuber **pictures** obtained from the **PHTS technology** will serve as a **training data** for a proof-of-concept of **prediction model development** for hyperspectral and RGB-based **early detection tool.**

For this, **IRIS** will be in charge to **build up** a multilayer digital tool (**smartphone or tablet app**) based on the pictures obtained from the PHTS technology which will act as a **storage unit** and **data processing unit** for the **image analysis (WP2)**.

The inserted data will be processed within the platform to generate an accurate response in terms of detection. The probabilistic machine learning models developed as part of the PhotonFood project will be used and extended in PataFEST (preliminary detection of mycotoxins and confirmatory software tool). In PataFEST, these methods will

[19] Buckseth, T., et al. (2016). Sci. Hort. 204: 79-87.
[20] Andrade-Piedra, J. L., et al. (2015). CIP, INIAP, CORPOICA.
[21] Rubio-Covarrubias, O. A., et al. (2017). Am. J. Potato Res. 94: 32-356.
[22] Rrashidi, M., et al. (2017). Plos One. 12(8): e0183283
[23] Prager, S.M, et al. (2022). Pest Manag. Sci. 78:3731-3745.
[24] Du, M., et al. (2012). Potato Res. 55:175–184.
[25] Carnegie, S., et al. (2003). Plant Pathology. 52(1):13-21.
[26] Munyaneza, J.E., et al. (2011). Am. J. Pot. Res. 88:435.
[27] De Diego, N., et al. (2017). Front. Plant Sci. 8: 1702.
[28] Ugena, L., et al. (2018). Front. Plant Sci. 9: 1327.
[29] Humplík, J.F., et al. (2015). Plant Methods. 11, 29.

allow for predictions of key traits based on genomic, environmental, and soil information, and allow for time series and missing data structures.

Complementary, an extra functionality will be **generated embedded** in the **platform** capable of **performing early detections** of the desired markers **(WP2)**. This functionality will be developed using **artificial intelligence techniques** by **FUND,** making use of the captured data. A priority will be to learn from data in image format, with the aim of providing the tool with a high level of usability using the smartphone or tablet camera. The necessary image pre-processing will be performed, and light computational cost algorithms based on deep learning techniques will be used, which can be applied from low-cost devices.

Furthermore, due to the **climate changes** and **globalisation scenarios forecasted**, the use of **AI** has been reported as a key **technology** to helping **save our environment** and significantly **reduce the ecological impact** in the production sector **(WP2)**. AI applies the power of machine learning to finding patterns in data that can help spot trends. One way to adopt a proactive attitude against the rise of temperatures in our continent is getting a profit of AI advantages to mitigate risks and adapt strategies as far as possible that offset the climate change. In this context, FUND will apply and **evaluate** different novel **predictive models** with the highest classification accuracy **base on AI techniques** as Convolutional Neural Networks (CNNs), Neural Network hyperspectral and Learning Vector Quantization (LVQ) algorithm-based method with the aim to detect and classify diseased potatoes and predict risks of pest disease spread due to climate change and globalisation.

Complementarily, **CRUN** will isolate and further identify *F. sambucinum* from the selected European potato candidate varieties (Table 1.1), which will be further used to inoculate tubers[30]. The **VOCs** related to potato dry rot will be **identified** using TD-GC-ToF-MS[31]. For this, the headspace of small potato microcosm experiment comparing diseased and non-diseased potatoes (specific pathogens will be inoculated for the different comparisons) will be analysed at different timepoints during the potato storage (monthly intervals for up to 6 months). This technique was chosen as it allows high concentration fold of the compounds produced (**quantification**) and hence **identification** of both, compounds with high concentration and those with very low concentration that are often missed/neglected when using solid phase micro extraction (SPME). A full metabolomic approach will be done using LC-MS/MS to identify non-volatile biomarkers, such as phenolic compounds, non-structural carbohydrates[32], and plant growth regulators[33], linked to the development (viz., incidence and severity) of the pathogens during potato postharvest storage. Spatial (*viz*., cortex, parenchyma) and temporal changes (monthly sampling throughout storage) in the biochemical composition will be determined per each treatment combination (viz., cultivar x fungal specie x season) to provide a better resolution of the impact of the disease progression in potato tubers. By this way, defining an early VOCs threshold will allow warehouses owners to detect in time the presence of pathogens through sensor devices and anticipate potato spoilage during the postharvest storage. For this, **FUND** will develop a **sensor array** for measuring the **concentration of those VOCs** by using **electrochemical sensors** based on MOFs[34], or on organic polymers, such as polythiophenes[35] **(WP2)**. These materials can be modified (doped) to improve their selectivity and provide accuracy of the electrochemical sensor. After that, several inks will be developed containing those modified materials, with other sensing compounds, organic solvents, surfactants, and binders and supported onto commercial electrodes by dip coating, spin coating, drop casting or inkjet printing. By this way, VOCs determination will be carried out electrochemically generating electrical signals enabling the monitorisation of its concentration. In case that individual sensors respond to a more than one selected biomarkers creating interferences, an e-nose could be developed[36] using recognition algorithms in order to improve the selectivity and accuracy of the sensors against VOCs concentration. IRIS will integrate the sensors developed by FUND into a printed circuit board (PCB) that will be able to **convert the electrical signal** coming **from the sensor array**, generating a **digital signal** compatible with the **user platform** (mobile app, e.g.).

<u>Pest and pathogen disease control solutions</u>

TECHN will develop a new family of **non-toxic** and sustainable **sprayable wax coatings**, in which bioactive natural compounds (e.g., essential oils) are embedded, to **provide** a **multifaceted protection** for **potato plant** leaves **against BC**, **CLso**, and **other plant pathogens (WP3).** Taking inspiration from plant cuticles that exhibit superhydrophobic and self-cleaning properties, the wax coatings will operate in both passive and active modes of protection. Coatings employ naturally derived fatty acids which self-assemble into crystalline structures with superhydrophobic properties, that reduce insect adhesion onto the treated leaves. TECHN has already demonstrated that the use of waxes can reduce the insect attachment forces on surfaces by up to 30-fold[37]. Many plant surfaces are known to be "slippery" (or anti-adhesive) to insects, in particular plants covered with nano- to micrometre-sized epicuticular wax crystals[38,39]. Thus, in PataFEST biomimetic coatings will form micro-rough surfaces (0.05-1 μm asperity size) which in turn

[30] Samson, R. A., et al. (2019). Westerdijk Fungal Biodiversity Institute.
[31] Gutierrez-Pozo (2021). PhD thesis, Cranfield University.
[32] Tosetti, R., et al. (2021). Posth. Biol. Technol. 173: 111420.
[33] Alamar, M. C., et al. (2021). Posth. Biol. Technol. 168: 111267.
[34] Li, H. Y., et al. (2020). Chem. Society Rev.49(17): 6364-6401.
[35] Li, Z., et al. (2019). Food Chem., 276: 291-297.
[36] ACS Sens. (2022), 7, 6: 1666–1675
[37] Gorb, E., et al. (2014). Beilstein Journal of Nanotechnology. 5(1): 1031-1041.
[38] Graf, C., et al. (2018). Bioinspiration & biomimetics. 13(5): 056010.
[39] Féat, A., et al. (2019). Progress in Organic Coatings 135: 331-344.

reduce insect attachment by its effect on both claws and adhesive pads. Moreover, coatings consist of natural, non-toxic waxes -fatty acids- which are known as natural antimicrobial agents[40], exhibiting intrinsic antimicrobial activity against numerous pathogens, and therefore serve as a second tier defence line against CLso and other relevant plant pathogens. Finally, coatings will contain small amounts of encapsulated essential oils (e.g., geraniol, thymol and their mixtures) which are slowly released to actively repel the vector psyllids. Several essential oils were already demonstrated repellence of adult BC and ability to reduce their nymphs in field experiments[41]. It should also be noted that the coating formulations combine abundant and non-toxic raw materials, being based on natural waxes and essential oils commonly occurring in nature, which together with the coating safe biodegradability, make them an environmental sound solution.

FUND will develop **biocontrol formulation prototypes** with **antifungal properties based on** the **action** of **biological catalysts** (enzymes) and **microorganisms** (yeasts) as an alternative of the chemical solution use (chlorine) to ensure the **soil-borne pathogen spoilage control** in **postharvest tubers (WP3)**. Biological control for postharvested fruits and vegetables is still considered incipient; however, its effectiveness against moulds and bacteria is well documented [42,43]. Most effective concentration of antagonist agents will be studied by in *in vitro* determinations.

To guarantee the viability of the application, distribution, uniformity, and adhesion of the antagonist agents on the potato surface, several water-based ingredients as gums waxes, vegetal oils and proteins will be used as carrier to formulate film forming coatings, provide barrier properties to gases and therefore reduce potato metabolism process and dehydration. Properties characterisation of the coating solutions as physicochemical, mechanical, gases barriers, microbiological and antioxidant properties will be assessed to achieve the best performance composition. Antifungal coating solution developed in PataFEST will represent a new economic and easy to apply technology to extend the shelf life of the potato during postharvest storage.

CRUN will work on **controlled atmosphere conditions** (CO_2 and O_2) during long-term postharvest storage. Tubers coated with biocontrol solution prototype will be subjected to different graduated CA environments with the aim to investigate the potential synergy of these technologies to reduce pathogen disease incidence and extend the postharvest shelf life of potato under commercial conditions. Along the storage period, metabolism of tubers, quality parameters, and presence of soil-borne pathogens will be evaluated.

Besides, **UPOL** will carry out simultaneous and cross-testing bioassays by the **application of PHTS** technology to **provide a selection of tolerance increasing formulations against** both **plant pest** and **postharvest pathogens** selected **(WP3)**. PHTS will be optimised by UPOL for a quantitative assessment of the efficiency of the formulations of the biocontrol coatings for postharvest potato tubers developed by **FUND** and testing their effect in the concentration/dose-dependent manner under various environmental scenarios simulating the possible postharvest conditions in the warehouses. Cross-testing using microtubers of different potato genotypes will allow defining potential genotype-dependent/specific response. The PHTS of **UPOL** will be further used for simultaneous evaluation of potential phytotoxicity of the wax sprays developed by **TECHN**. A variant of PHTS employing model plant Arabidopsis will be used throughout the development of the spraying wax formulation to exclude in the early stages of the formulation development such components that can cause phytotoxic effects on treated plants. The advantage of PHTS is the possibility of testing large numbers of components in various combinations and concentration ranges in one run. This will help to significantly speed up the process of the spraying wax formulation development and allow precise estimation of the safety ranges for every experimental formulation. High-precision phenotyping using RGB cameras, FluorCam, IR, and hyperspectral cameras will be further used to validate the phytotoxic safety of the final formulation(s) and for detailed description of the effects of the application on the physiological status, growth and development of targeting plants and organs.

Soil pest management strategies

Innovative and sustainable **soil-applied solutions** will be explored by UNIBO for enhancing the potato plant health status and therefore decreasing the crop susceptibility to field and post-harvest pathogens **(WP3)**. To do that, biomasses residues will be used to obtain carriers able to functionalise the plant availability of several nutrient elements such as boron (B), copper (Cu), iron (Fe), manganese (Mn) and zinc (Zn) which play important physiological roles on potato plant health reducing susceptibility to pathogens of both foliar and tuber. The laboratory production of functionalised material is a crucial point of the action therefore the efficiency of the functionalisation needs to be well tested under lab conditions before performing in *in vivo* trials, to ensure on one hand the stability of nutritional elements on the carrier and, on the other hand, the efficiency of the carrier to release the same elements and ensure their availability for the plant. Moreover, the innovative fertilisation will be tested under CLso infection to assess the effect of the most performing fertilizers application protocol(s) on the potato plant health status derived from the optimised nutrient availability in the soil. The pathogen on its potato host will be introduced in a dedicated section of the greenhouse according to the EU plant health legislation and all experimental trials will be conducted

[40] Desbois, A. P., Smith, V. J. (2010). App. Microbiol. Biotechnol. 85(6): 1629-1642.
[41] Yang, X.B., et al. (2010). Crop Prot. 29: 1320-1324.
[42] Cissé, M., et al. (2015). Posth. Biol. Technol. 101: 10-14.
[43] Aloui, H., et al. (2015). Int. J. Food Microl. 200: 22-30.

under the surveillance of the national and local (Emilia-Romagna region) Plant Protection Service (Fig. 1.8). Trials involving CLSo will be exclusively planned under controlled conditions to ensure the pathogen confinement.

Figure 1.8. Plant Pathology Greenhouse (UNIBO facilities)

To measure the effects of the new fertilisers application, the level of CLSo in potato tissues (leaves and tubers) will be also assessed by a precise molecular tool such as Droplet Digital PCR (ddPCR) coupled or not to Reverse-Transcriptase (RT) reaction in order to evaluate changes in the infection level (ddPCR) and in the biological activity (RT-ddPCR) of the pathogen. The effect of the improved potato plant health on reducing the susceptibility to postharvest tuber pathogens will be also evaluated by specific experiments under standard storage conditions and artificial inoculation to ensure the adequate normalisation of trial conditions. This alternative and sustainable technology will allow the reduction of chemicals used to control CLso vector and fungal pathogens of storage, as well to improve the nutritional properties of the tuber enriched by microelements.

Besides, **soil disease incident** trials under **different climate scenarios (WP3)** mimicking predicted drought periods will be **conducted** at **CRUN in a phenotyping glasshouse** facility designed to trial crop-soil systems at pilot scale under a multi-sensor gantry as described above (Fig. 1.7). Sensors on the phenotyping platform will be used to evaluate if signals can be identified to detect drought stress and how these interact with disease stress by comparing inoculated and no-inoculated lysimeters. The facility replicates all stages of the whole crop production cycle plus rotational context (multiple cropping seasons) with predefined soil physical conditions affecting fungal and bacterial movement, and recreates the dynamic interrelationships of soil health, water use, biotic stresses (pathogens, weeds, pests), environmental variables.

Integrated plant management

Several IPM strategies will be considered in PataFEST project testing previously identified resistance potato varieties; early detection and surveillance tools (mobile app and VOCs sensor); and natural sustainable solutions for plant, tubers and soil manufactured by CHEM **(WP4)**.

In Ecuador, one IPM strategy will be carried out through **field bioassay** trials in INIAP facilities under real conditions (presence of CLso and its vector in potato plant) evaluating the behaviour of selected potato varieties for their resistant traits against CLso and its vector. Besides, INIAP will validate the capacity of the **image analysis digital tool** (mobile app) for early detection and surveillance of CLso disease in potato plants in fields and harvested tubers. Afterwards, digital tool will be also validated in potato plants and tubers infected with ZC under field conditions [44,45] in representative places with high pressure of the disease in the Ecuadorian highlands.[46,47]

Finally, **pesticide-free natural solution** performance will be evaluated on potato plants to study the repellent effectiveness against BC[48]. The natural solutions with better control of BC will be validated under field conditions using large plots (at least 2 ha) and the antifungal properties against soil-borne pathogen in harvested tubers.

In Europe, one IPM strategy will be organised in potato warehouses of TERREA (France), BOEH (Germany) and AB (UK) to validate postharvest technological solutions. **Biocontrol** superficial **coating** solution, **image analysis mobile app** and **VOCs sensor** will be tested to early detect and prevent the pathogen spoilage in potato. At least **1 ton of tubers** (two ware potato varieties selected including organic crops) will be coated by biocontrol coating prototype and **100Kg of tubers** will be used to validate the imagen analysis tool and VOCs sensor. Periodically, the effectiveness of biocontrol coatings will be assessed by determining the incidence and severity of soil-borne pathogens during the storage period established.

Besides, **a second IPM strategy** in Europe will be organised in UDAPA (Spain), BOEH (Germany) and TERREA (France) potato fields where innovative and sustainable **soil solutions** will be tested for enhancing the potato plant health and pesticide-free natural solution performance will be evaluated on potato plants to study the repellent effectiveness against aphids. Application will be implemented in **1 ha** under real cultivation conditions and treatments effect will be assessed by determining the soil quality and presence of aphids in potato plant treated.Application will be implemented in **1 ha** under real cultivation conditions and treatments effect will be assessed by determining the soil quality and presence of aphids in treated potato plant.

Citizen science

ECSA, ANEC, KNEIA and FUND will be in charge of citizen science activities that will be organised along the PataFEST project execution. Results obtained will significantly contribute to boost the collaborative efforts to detect and control emerging pest threat and pathogens disease incidence. These activities consist of **co-creative workshops**

[44] Garhwal A.S., et al. (2020). Biosystems engineering. 197:306-317.
[45] Bienkowski D. et al. (2019). Computers and Electronics in Agricul. 167, 105056.
[46] Garhwal A.S., et al. (2020). Biosystems engineering. 197:306-317.
[47] Bienkowski D. et al. (2019). Computers and Electronics in Agricul. 167, 105056.
[48] Jorgensen, N. (2013). New Zealand Plant Prot.66:333-340.

(**WP 1 and 4**) in different geographical locations as 1) Berlin-Bradenburg, 2) experimental field station in Spain and 3) potato fields of small-scale farmers in Ecuador. The aim of these workshops will be to inform the participants about the PataFEST project, promote scientific experiment demonstrations and discussion with PataFEST scientists, and share concerns and questions related the integration of strategies to overcome emerging crops pest and disease threats. The workshop activities will be addressed to a broad category of citizen. Workshops based on **molecular biology experiments** involving citizens of all backgrounds with a strong interest in science will be held in the "Long Night of Science in Berlin", and another one in the "The Doctor's Farm, Experimental Field Station" of ANEC in Spain. The workshop activities will be addressed to a broad category of citizen. Workshops based on molecular biology experiments involving citizens of all backgrounds with a strong interest in science will be held in the "Long Night of Science in Berlin", and another one in the "The Doctor's Farm, Experimental Field Station" of ANEC in Spain. In Ecuador field trials activities will allow potato farmers to carry out free trials of a selected biorational insecticides solution on potato plants and evaluate the repellent effectiveness against BC. In Europe, urban garden owners in Berlin-Bradenburg region and small horticulture crop farmers in Spain will have the opportunity to carry out free trails of natural solutions for plant and soil.

As part of the communication and dissemination activities, ECSA will organize webinars addressed at different target groups and with different aims, ranging from introducing PataFEST project to the citizen science community, to share more technical information for the agriculture community. ECSA's working group on Citizen Science and Agrifood will be a supporter of this activity.

Joint Research Center involvement (WP 4)

PataFEST project will envisage the involvement of the **Joint Research Center (JRC)** to develop reference materials in support of a harmonised IPM in the EU legislative acts and policy recommendations. Besides, JRC will be tasked to support AINIA in the elaboration of Life Cycle Analysis assessments. JRC activities will be more focused on the social life cycle assessment (S-LCA) evaluating the real and potential social and socio-economic impacts of PataFEST implementation strategies within main important potato stakeholder groups as farmers, producers, traders and consumers.

1.2.2 Related research and innovation activities

The activities developed in PataFEST linked with many previous and ongoing R&D initiatives described below:

Project	Main result/knowledge feeding into PataFEST
PAPACLIMA (FAO, 2016-2019). Partner: **NEIK**	Detection of candidate genes against fungi. Knowledge and software for performing different association mapping techniques. Evaluation of fungal diseases.
SPIRIT (ACP-EU Coop. Programme S&TII, 2012-2016). Partner: **NEIK**	Analysis of the potato value chains in East African countries (literature research, on-site visits and meetings with farmers, research institutions and stakeholders, information compilation, database establishment, Strategy Development).
POISE (EIG concert Japan, 2016). Partner: **UPOL**	Identification of biomarkers related to tolerance level to abiotic stress in tomato and rice plant.
QCAP: (Interreg NWE, 2017-2019). Partner: **CRUN**	Detection and identification of potato spoilage VOC biomarkers under postharvest cold storage conditions.
PHOTONFOOD (H2020, 2021-2024). Partner: **IRIS**	Development of the global system control, software and data management and processing platform.
ANTISEPTIC-WAX (ERC H2020-PoC, 2018-2020). Partner: **TECHN**	Environmentally friendly wax-spray green technology to eliminate attachment and development of pests on plant leaves surfaces.
REPEAR (FP7, 2014-2017). Partner: **FUND**	Development of superficial coatings based on natural additive to reduce respiration rate, control dehydration, and pathogens spoilage of postharvest pears.
ADAPT (H2020, 2020-2024). Partner: **EUPAT**	Dissemination of the project objectives and results via internal and external communication channels.
TOXDTECT (FP7, 2013-2016). Partner: **FUND**	Development of array of sensors embedded within the meat package to determine the real state and properties of the meat.
BIOMUSH (FP7, 2012-2015). Partner: **FUND**	Development of a tool for early detection of mushroom fungal diseases.
FF-IPM (H2020, 2019-2023). Partner: **ANEC**	Pest prevention and off season focused IPM against new and emerging fruit flies

1.2.3 Gender dimension

PataFEST does not include the integration of the **gender dimension (sex and gender analysis) in R&I as the topic 2022-F2F-02-02 does not require it**. Nevertheless, the project has a strong multi-actor participatory approach and involvement of citizen, thus it will commit to address the diverse needs of citizens regardless their gender and sex.

1.2.4 Open science practices

The project will follow an Open Science Strategy (OSS). The preliminary type of research data, exploitation route and IP strategy will be outlined in the Exploitation Plan (WP5). This plan will show that some results with commercial and/or industrial important value will be protected as IPR or trade secrets, and thus they will not be included in PataFEST's OSS. The research results obtained by the work carried out by researchers will follow these Open Science practices: **1) Open Access (OA) to peer-reviewed scientific publications** ensuring at least under upon publication, deposition of the author accepted manuscript (AAMs) or version of record (VOR) in a **trusted repository** and immediate open access via the repository under CC BY or equivalent. Alternatively, some research results will be made available in the **Open Research Europe**; 2) **Early and open sharing of research** through pre-registration, registered reports, pre-prints or crowd-sourcing; 3) Researchers will participate as much as possible in **open peer-review** in related topic of PataFEST; **4) Multi-actor approach activities** in the co-creation of tasks from the beginning and across the project duration.

1.2.5 Research data management and other research outputs

The Data Management Plan (DMP) will consider all the data that the project will generate and use and will detail whether and how it will be used, exploited or made accessible for verification and re-use; how and under what conditions it will be curated; and how it will be interoperable with regard to specific quality standards. Data refers to any kind of result including statistics, results of experiments, measurements, observations resulting from trials and demonstration, survey results, interview recordings, images and bibliographic metadata used for the validation publications. According to EC recommendation, as part of making data findable, accessible, interoperable and re-usable (FAIR), the DMP will include information on: **1) handling of data during & after** the end of PataFEST project; **2) what data** will be collected, processed and/or generated; **3)** which **methodology and standards** will be applied; **4)** whether data will be **shared/made in OA**; **5)** how data will be **curated and preserved** (including post-project). DMP will not have impact on IPR, because the decision to publish OA documents come after a more general decision on whether to first seek protection for IPR. The data will be deposited via OA in a trusted repository under the latest available version of the Creative Commons Attribution International Public License (CCBY) or Creative Commons Public Domain Dedication (CC0) or a licence with equivalent rights.

1.2.6 Compliance with the ´do no significant harm principle (DNSH)'

All the research activities of PataFEST will comply with DNSH principle under the Recovery and Resilience Facility Regulation of EU. PataFEST fulfils DNSH in different ways: 1) no activity is harming the **climate change mitigation** as the developed plant health products are fossil-free; 2) no activity is harming the **circular economy** concept. All the contrary, the project will apply circularity principles in the plant health protection products leading to more resource efficiency, to zero-waste practices, and to sustainable and 4) no activity is harming **biodiversity and ecosystems** as the project enhances potato genetic variety on agricultural land.

2 IMPACT

2.1 Project's pathways towards impact

This section covers **PataFEST's pathways towards the Expected Outcomes (EOs)** defined in the topic at medium-term (from project's end to 4 years later) and these two **Expected Impacts (EI)** of the Destination 'Fair, healthy and environment-friendly food systems from primary production to consumption' of the Cluster 6 Work Programme:

- **EI 1**: sustainable, productive, climate-neutral and resilient farming systems […], while minimising pressure on ecosystems, restoring and enhancing biodiversity, […] generating fair economic returns for farmers.
- **EI 3**: sustainable, healthy and inclusive food systems delivering co-benefits for climate mitigation and adaptation, […], sustainable healthy nutrition, safe food consumption, food poverty reduction, the inclusion of marginalised people, the empowerment of communities, and flourishing businesses.

These pathways are defined by clear, measurable, quantified, and assessable **impact indicators** that fall under three **impact categories**: **scientific, economic & technological and societal impacts.** The results of the project will be used or further up-take by direct **target groups**. PataFEST will generate credible and sustainable impact pathways mainly in these 5 impact areas set by the European Commission: 1) *Sustainable food systems from farm to fork on land*, 2) *Enhancing ecosystems and biodiversity on land, 3) Clean Soil*, 4) *A resilient EU prepared for emerging threats* and 5) *Inclusive growth and new job opportunities* in line with some of the 9 **Key Impact Pathways** (**KIPs**) defined within the **Key Strategic Orientation KSO B** '*Restoring Europe's ecosystems and biodiversity, and managing sustainably natural resources to ensure food security and a clean and healthy environment'* of the HEU Strategic Plan.

SCIENTIFIC IMPACT

Increase the information of the molecular and biological potato RGs and ecological analysis of its pest (CLso) and vector (BC). *Corresponding to project´s objective O1.*

- *Impact Indicator*: Acquisition of data related to potato RGs.
- *Target groups*: Scientific/Technical community, farmers, industry along the potato value chain, plant health policy makers.
- *Contributions to Key Impact Pathways:* KIP 1 'Creating high-quality new knowledge'.

Zebra chip disease is both costly and difficult to manage. Application of chemical pesticides has been the primary method of combating ZC in numerous regions around the world. Unfortunately, overuse of pesticides has resulted in ZC-vector resistance. As a result, pesticide use is both economically and environmentally unsustainable, and goes against the **2030-EU Action Plan** target of lowering chemical pesticide use by 50% to achieve the "zero pollution" targets. PataFEST proposes a method for studying potato resistance genes to the ZC vector psyllid BC and/or CLso based on potato genetic resources. This will ensure a long-term sustainable and successful strategy for managing increasing ZC disease and cultivating more resistant potatoes. PataFEST will thus offer a strategy in line with the **EU Biodiversity Strategy 2030's** primary action to enhance genetic variety on agricultural land, which would benefit EU health and economy. To achieve this impact, the project proposed the following pathway:

Expected Outcome in the topic	Project Contribution	Scale & Significance
EO 1: Find adequate responses to new and/or emerging plant pests	***Short term:*** Collection of all the available genetic information of BC and CLso, and identification and characterisation of potato RGs against diseases caused by ZC and soil-borne pathogens. ***Medium term:*** Complete map of potato genes resistance to emerging threats in EU.	***Linked to results R1-R4*** •Molecular monitoring protocol of pest spread (CLso and its vector) pathway across different Ecuadorian regions. •500 potato samples analysed: 355 European samples (40 wild species, 35 native varieties, 280 old and new cultivars) and 150 samples from Ecuador.
Expected Impact in the Destination	**Project Contribution**	**Scale & Significance**
EI 3: delivering co-benefits for climate mitigation and adaptation	***Long term:*** Prevent similar losses in potato production in the EU as those seen in other countries where these pests already exist. Similar molecular characterisation strategies could be replicated to other type of crops (e.g. carrot, pepper, etc).	At least 5% (of the above 500 samples of potato resistance varieties for targeted diseases and pests will reach the market.

ECONOMIC/TECHNOLOGICAL IMPACT

Long-term sustainable potato pest and postharvest diseases integrated management, including prevention, early detection and surveillance strategies. *Corresponding to project´s objectives O2 and O3*

- *Impact Indicator*: Catalogue of innovative solutions to respond to new and emerging potato pests.
- *Target groups*: Farmers, Bio-based industry of plant health solutions, Industry along the potato value chain, Citizens.
- *Contributions to Key Impact Pathways:* KIP 7 'Generating innovation-based growth', KIP 9 'Leveraging investment in research and innovation'

PataFEST's concept is structured around the **EU's agricultural research and innovation strategy**, notably the set of initiatives aimed at "**Priority 2: healthier plants**"[49]. As the EU strives to reduce the use of chemical pesticides[50], **biopesticides are being more widely used in agricultural practices** (biopesticides CAGR 20.0% versus synthetic pesticides CAGR 4.7% for the period 2018-2023; Fig. 2.1), potentially introducing new pathogens not occurring before[51]. PataFEST will develop sustainable and systemic IPM strategies to address the prevention, monitoring, control, and management of ZC during cultivation and soil-borne pathogens during postharvest storage, as an alternative to the use of pesticides or other contentious methods, in combination with other innovative agronomic and digital solutions. In this regard, the project is fully in accordance with the actions of the **"Agriculture and the Green Deal"**, as it promotes the adoption of a long-term IPM by potato professionals to achieve a healthy and sustainable food system for people and the earth[52]. To achieve this impact, the project proposed the following pathway:

[49] A strategic approach to EU agricultural research & innovation (2016). Final Paper

[50] Directive 2009/128/EC. Sustainable use of pesticides. Available here.

[51] Frost & Sullivan (2019). Global Crop Protection Market, Forecast to 2023.

[52] Integrated Pest Management (IPM) in the EU Green Deal. Available here.

Expected Outcome in the topic	Project Contribution	Means of Verification
EO 2: Understanding of drivers of plant pest emergence, influence of climate change, ecosystem degradation and globalisation *EO 3: economic and environmentally sound solutions for effective pest management in farming*	***Short term*:** integrate sustainable innovative solutions and validate them as an Integrated Pest Management strategy at TRL 5 in 5 different geographical areas (**Germany, France, Spain, UK** and **Ecuador**) based on their respective climate conditions and globalisation factors. The technologies are: a)Low-cost, user friendly and robust portable sensors and digital tool (smartphone app) (**R5**). b)Prediction models based on AI algorithms (**R6**) c)Development of bio-based sustainable solutions applied to potato plant, tuber and soil (**R7-R9**) and validation in field trials and warehouses (**R11-R13**). d)Bio-assay field trials of selected potato varieties affected by the bacteria in Ecuador (**R10**). ***Medium term*:** laying the foundations for future development (TRL 5→8) of novel potato diseases strategies (e.g., bio-based chemical products stimulating the expressing of RGs).	***Linked to results R5-R13*** • Reduction of the presence of the bacteria (BC) by 50%. • Reduction of the incidence of ZC by 25%. • Reduction of the incidence of soil-borne pathogens by 40-50%. • VOCs sensors tested in 3 potato warehouses (100 Kg of potato/each). • User friendly mobile hardware to detect CLso with 90% of reliability in 30 potato varieties of Ecuador.
Expected Impact in the Destination	**Project Contribution**	**Means of Verification**
EI 1: sustainable, productive, climate-neutral and resilient farming systems […],minimising pressure on ecosystems, restoring and enhancing biodiversity,[…] fair economic returns for farmers.	Exploitation and dissemination of a rigorous, sustainable and economic Integrated Pest Management to control emerging threats and diseases for potato crops and other crops in Europe. The project envisaged the collaboration with the **Joint Research Center** in order to develop reference materials in support of a harmonised IPM in the EU legislative acts and standards.[53]	Publication of EU guidelines and/or action plans to prevent potato diseases based on PataFEST's IPM strategy. EU countries primarily benefited will be the main potato producers (Fig 2.2).

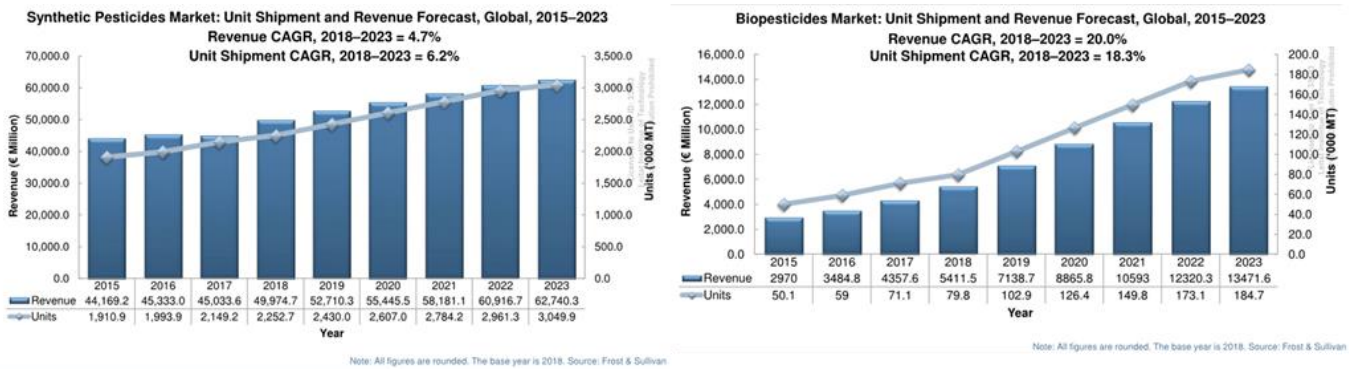

Figure 2.1. Growth opportunities of synthetic pesticides versus bio-based pesticides. Source: Frost & Sullivan (2019). Global Crop Protection Market, Forecast to 2023.

SOCIETAL IMPACT - 1
Citizens and farmers as key actors for prevention and early detection of potential potato threats. *Corresponding to project´s objectives O4.* • *Impact Indicator*: Involvement of citizens across regions • *Target groups*: Citizens, Farmers, Producers. • *Contributions to Key Impact Pathways:* KIP 6 'Strengthening the uptake of research and innovation in society'
The potato IPM proposed in PataFEST will be **designed by and for citizen, farmers and potato producers in the EU and Associated Countries**. Citizens will contribute to the studies in a variety of ways through citizen science activities (engagement campaigns, workshops to collect scientific and technical data, participation in scientific webinars) inspired by other EU initiatives such as *eu-citizen.science* or the international citizen hub *'Science we can do together'*, with the goal of preventing and detecting potential threats to potato crops around the world and long-term adoption of the IPM. Everyone, regardless of age or professional background, will contribute to the

[53] JRC: Reference & Measurements. Available here.

project's research outcomes in order to have a significant impact not only during the project's implementation (2023-2026), but also in the medium and long term. The project will ensure low-cost and effective means of maximizing our outcomes across regions of the EU and internationally by involving citizens in the project from the beginning, throughout its duration, and beyond. Citizens will play a significant role in supplementing knowledge to technical pest management strategies. To achieve this impact, the project proposed the following pathway:

Expected Outcome in the topic	Project Contribution	Means of Verification
EO 4: Support to relevant plant health policies in the EU and Associated Countries.	**Short term**: Scientific and technical data are collected and supported by the citizens across the project to monitor emerging pest. **Medium term:** Share best-practices of the citizen science approach to other related pests.	***Linked to results R14-R15*** Three types of citizen's activities will generate the necessary scientific data and dissemination through the webinars. • 2 molecular biology workshops (Spain & Germany): 30 persons/each (WP1). • 3 co-creative workshops (Spain, Germany & Ecuador): 40 persons/each (WP 4). • 8 citizen science webinars: 50 participants/each (WP 5).
Expected Impact in the Destination	**Project Contribution**	**Means of Verification**
EI 3:[…]the inclusion of marginalised people, the empowerment of communities, and flourishing businesses	Adoption of citizen science as a tool for the surveillance and control of disease-carrying vectors by governments and private sector.	At EU level, citizen of science is incorporated into the main research agendas and policies to protect European plant diversity and health.

SOCIETAL IMPACT - 2

Creation of an interactive innovation model to help the implementation of potato IPM across different professional backgrounds at local, regional, national and international scale ('Multi-Actor Approach'). *Corresponding to project´s objectives O5.*

- *Impact Indicator*: Practice abstracts of EIP-AGRI
- *Target groups*: All actors of the potato value chain shown in Fig. 2.1.
- *Contributions to Key Impact Pathways:* KIP 4 'Addressing EU policy priorities and global challenges through R&D' KIP 5 'Delivering benefits and impact through research and innovation missions'

The EIP-AGRI policy goal of "speeding up" innovation will be met from the start of the project and in the long run thanks to the active participation of farmers and agri-food businesses through the key associations and cooperatives engaged (ANEC, UDAPA, EUPAT). The multi-actor participatory platform will include the scientific & technical community, companies (SMEs, large industries, agricultural/potato associations, and cooperatives), and citizens/consumers across the potato value chain (Fig. 2.1) to share knowledge and cooperate one with each other to maximize our results and outcomes. This platform will be critical for enhancing rural community engagement at all scales (local, national, and international) and for disseminating the project's proposed solutions against emerging potato pests. PataFEST's impact will be amplified by close engagement with current EIP Operational Groups (via ANEC) and other similar projects.

Expected Outcome in the topic	Project Contribution	Means of Verification
EO 3: economic and environmentally sound solutions for effective pest management in farming *EO 4: Support to relevant plant health policies in the EU and Associated Countries.*	**Short term:** Share knowledge, experience, and background due to the complementary disciplines of the multiple actors **Medium term**: acceptance and adoption of the PataFEST solutions by farmers and other actors of the potato value chain production.	***Linked to results R16-R18*** •Multi-actor participatory platform (4 years + 5 additional years after projects end). •60 **"practice abstracts"** for EIP-AGRI •2 intermediate workshops including policy discussions: 60 participants in each. •1 Final Conference (M48): 120 participants. •1 Policy Recommendation Report (M40)
Expected Impact in the Destination	**Project Contribution**	**Means of Verification**
EI 1: sustainable, productive, climate-neutral and resilient farming systems	Implementation of potato IPM across all the actors and main EU potato producers	Farmers, potato producers and traders are aware of the potential threats and they are

[…]. EI 3: sustainable, healthy and inclusive food systems delivering co-benefits for climate mitigation and adaptation […]	→ ensure economic sustainability of potato EU production (Fig 2.1), increase farm competitiveness and improve key actor's position in the value chain.	also well informed and trained of the strategies to carry in case of potential impact in their lands and production.

All in all, PataFEST is built up with a strong commitment towards the key policies objectives of the **new Common Agricultural Policy (CAP)**, **Farm-to-Fork strategy** and **SDG (2, 8, 13, 15, 17)** for the period 2023-2027, briefly: **1)** "Fostering knowledge & Innovations" → involving farmers to our research activities and outcomes**; 2)** "Halting & reversing biodiversity loss" → preserving more resistance potato varieties, **3)** "Efficient natural resource management" → developing a sustainable, effective and long-term management strategies to tackle future potato pest; **4)** "Contributing to climate change mitigation" → boosting sustainable and bio-based circular solutions for plant health (biopesticides from biomass sources); **5)** "Increasing competitiveness" → involving all relevant actors in the potato value chain. The project also will support the "**Plant Health Law**" and all the associated implementing acts related to pests, commodities, plant health requirements, etc[54].

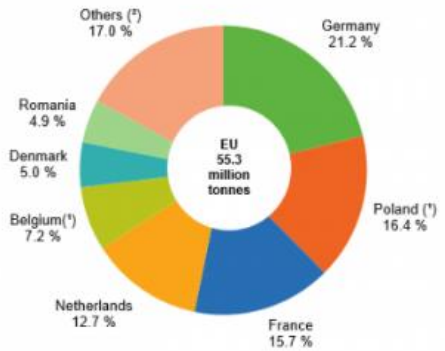

Figure 2.2. Production of potatoes, including seed potatos, by EU countries, 2020 (UK is excluded: its production was 5.2 mtonnes, 2019). [3]

Requirements and potential barriers

Barrier/Requirement Description	Strategies for overcoming
Technology demand: low demand of the potato industry and/or bio-based industry of plant health solutions could undermine the further development of the solution at higher TRLs.	Scientific/Technical community of the project, in collaboration with other funded EU projects, initiatives and OPs of EIP-AGRI will create dedicated scientific forums to mitigate this potential barrier.
Lack of **citizen engagement** in the citizen science tool and poor adaption by farmers of IPM.	PataFEST will provide well documented results, dissemination materials and engagements activities to all the actors that will form a sound base to support our results and outcomes.
EPPO Standards associated to potato pests & guidelines[55]: PM1/002(30) EPPO A1 and A2 Lists of pests recommended for regulation as quarantine pests (2021); PM7/143(1) CLso; PM8/001(2) Potato; PM9/025(2) BC and CLso and the guidelines in good plant protection practice – Potato PP 2/2.	The novel potato management strategies for both pre- and postharvest will be carefully followed these standards in order to avoid any potential confrontation with existing standards.
EU Regulatory and Policy requirements: EU regulatory framework may affect production and commercialisation of potato pest management solutions. The Plant Health Regulation and EU Directive 2002/56/EC are key for the correct use of IPM to protect potato crops beyond project's duration.	The consortium, specifically our partners representing different actors across the potato value chain, will review the legislation applicable to protective measures against pests and other legislations applicable to PataFEST. Moreover, engagement with policy makers will be ensured in the two policy events organised within the project's workshops.

2.2 Measures to maximise impact - Dissemination, exploitation, and communication

PDEC (Plan for the Dissemination, Exploitation and Communication activities): This section includes a first version of the PDEC, including all the measures to maximize the expected outcomes and impacts of PataFEST. The purpose of our PDEC (**M6)** is to provide a formal planning document for using, communicating and disseminating knowledge throughout the project, and to ensure that as many stakeholders as possible have access to these results. This includes: the potential end-users and relevant industrial sectors, financial and market actors, environmental consultancies, policy makers, scientific community, mass media and multipliers for addressing the general public such as consumer associations, as well as other targeted stakeholders (EC and national R&D initiatives). The most appropriate communication and dissemination tools and channels will be selected for each target audiences identified. The plan will assure the dissemination does not interfere with the IPR management but serve it. To protect the exploitation potential, a **confidentiality policy will underpin our dissemination activities:** all information is treated as confidential unless otherwise agreed. The PDEC will be developed under the supervision of the DCE

[54] Plant Health Law (Regulation (EU) 2016/2031 and implementation acts are found here.

[55] All the documents related to EPPO standards for potato are available here.

Manager (KNEIA), to ensure a high visibility of the funded actions and to contribute to the expected outcomes and impacts as described in 2.1, allowing also **future exploitation** of PataFEST results. The PDEC will define: the purpose of the dissemination and communication actions; the message; the type of audience; the communication methods; the timing over the project execution. Performance indicators will be established as part of the plan, such as number of citations, number of hits on web, number of downloads, number of industry consultations (from tradeshows/exhibitions), number of policy-maker engagements, etc. **Here below the main PDEC's elements are described in terms of Dissemination, Communication and Exploitation**.

Publication of research result: PataFEST expects to publish several research papers detailing our scientific advances. The dissemination activities will be carried out through **open access publications** in **specialised journals** (*Journal of Plant Diseases and Protection, Journal of Plant Protection Research, Plant Health, Plant Health Progress, Potato Journal, American Journal of Potato Research, Potato Research, etc.*) and **Open Research Europe**. All these publications reach a wide spectrum both of the scientific, public administration and technical community. Other channels such as the **Horizon Results Platform**, the **Innovation Radar**, and other EC publications will be targeted. PATAFEST will develop **60 "practice abstracts"** to contribute to EIP-AGRI repository.
Target Audience: *Research and academic stakeholders; Standardisation bodies and policy stakeholders, national authorities, EC, national governmental bodies, General public and Mass media*
Key Quantified targets: ***Scientific publications*** *= ≥6/8 articles published in Q1 journals or magazines;* ***EIP-AGRI practice abstracts*** *= ≥60;*

Presentation at conferences, events and trade shows: Key to our dissemination strategy is the **"PataFEST Roadshow**", an agenda of events that will be mapped out for the duration of the project. To maximise resources and bring mutual benefit to events organised by third parties, PataFEST will schedule out the trade fairs, exhibitions, workshops, seminars, conferences, etc. of relevance in Europe and internationally that consortium members can attend. Our target is to have PataFEST present in **at least 24 events in Europe and internationally** during the project -**Conferences** and **Industrial Fairs**: *International Plant Health Conference, Conference of the European Foundation for Plant Pathology, International Conference on Environmental Science and Applications, International Conference on Global Plant Health Assessment, Global Summit on Plant Science*. A philosophy of 'not wasting' any dissemination opportunities will be instilled in the consortium encouraging partners to capitalise on the relevant marketing and dissemination events they are attending or organising to plan a dissemination activity for PataFEST. Finally, PataFEST will take advantage of dissemination events organised by the EC to present the project results as for example **Horizon Europe Info Days**.
Target Audience: *Farmers, Industry along the potato value chain, Bio-based industry of plant health solutions, Industry Multipliers-Associations, Innovation programs, Policy stakeholders, national authorities, EC, national governmental bodies;*
Key Quantified targets: ***Conferences and events*** *= ≥24 events by the end of the project;* ***Horizon Impact Award*** =≥1 application; ***Conference publication*** *= ≥ 1 conference proceeding accepted per year (starting in 2nd year) per research organisation (27 total);*

Dedicated dissemination events: PataFEST plan to organize the final conference event in M48 (KNEIA and FUND) to release its results to all key stakeholders and maximize the project's impact. A policy makers' session will be organis
ed, inviting members of the Commission expert group on Plant Health Legislation. Key aspects of the event will be accessible online to allow stakeholders unable to travel to participate. In addition, KNEIA and FUND will organize two intermediate workshops on the project execution and preliminary results together with brother/related projects and clusters to increase the project's visibility and reach.
Target Audience: *Farmers, Industry along the potato value chain, Bio-based industry of plant health solutions, Industry Multipliers- Associations, Innovation programs, Policy stakeholders, national authorities, EC, national governmental bodies;*
Key Quantified targets: ***2 Intermediate Workshops*** *= ≥60 participants in each;* ***M48 Final Conference*** *= ≥120 participants*

Online dissemination & communication: PataFEST will launch a dedicated **website on M6**, populating it with regular non-confidential updates on our progress and a real-time "scoreboard" for our indicators. The web will serve as an enduring internet site for promoting the project's results and housing all the downloadable public deliverables and outputs of the project, and it will also house our '**PataFEST Green Channel' for open access** to the peer-reviewed publications of the project. The website will be supported by other instruments like the project **corporate identity tool**, the **newsletter** and **e-bulletins**. In addition, on M30 the project will launch the Multi-Actor Participatory Platform that will last at least 10 years after the project end (exploited by KNEIA). Each partner will leverage their own websites and other online presence (social media, blogs, newsletters and on-line platforms) to disseminate

information about the project, increase visibility of the project's website and publish **press releases** stating the project objectives in each participant state to drive early adoption and interest. PataFEST will also use **social media** (**Twitter and Linkedin**) and dissemination channels to maintain the momentum until preliminary results can be published. **Two videos –** an introductory video on M6 for the general public and a divulgative report video on M48 describing the final results- will be developed and published on YouTube, project website and social media, to raise awareness and engage stakeholders with the project activities and to promote uptake during/after the project. In addition, the project will make use of any dissemination route offered by the EC, including publication of our project briefings on CORDIS and similar websites (**CORdIs News, Cordis Wire, Research*eu results magazine, Research*eu focus and Openaire)**.

Target Audience: *Farmers, Industry along the potato value chain, Bio-based industry of plant health solutions, Industry Multipliers- Sectoral Associations, Environmental Consulting & Engineering, Innovation programs, General public and Mass media*

Key Quantified targets: ***Corporate identity: logo and templates =*** *Used in all deliverables / materials;* ***Newsletters =*** *≥ 200 subscribers by the end of the project;* ***Project website*** *= ≥8000 visits;* ***Project Videos*** *=≥ 2 videos;* ***Press Releases*** *= Direct engagements with 5 journalists and/or influencers;* ***Twitter*** *= ≥ 400 followers;* ***Linkedin*** *= ≥ 200 followers by the end of the project.*

Off-line dissemination & communication: A set of off-line dissemination tools will be developed such as the project **leaflets, infographics, factsheets,** posters and roll-up. They will contain different information depending on the information to be exposed. The leaflet will be promoted in fairs and congresses as well as in everyday business life of all partners. Furthermore, potentially specific leaflets or **posters** for technical activities /technology will be developed to expose the main research and development activities to involve specific stakeholders in the project. Moreover, a **roll up** will be developed to be used in each event to make visible PataFEST general concept.

Target Audience: *Farmers, Industry along the potato value chain, Bio-based industry of plant health solutions, Industry Multipliers- Sectoral Associations, Environmental Consulting & Engineering, Innovation programs*

Key Quantified targets: ***Infographics & factsheets =*** *≥ 200 views by the end of the project*; ***Leaflets*** *= ≥1000 leaflets distributed/organisation reached;* ***Project Roll up =*** *≥5 events where the roll-up is displayed;* ***Posters*** *= ≥ 2 posters*

Policy Feedback: PataFEST will implement several direct and indirect dissemination measures to contribute to policy shaping and supporting the implementation of new policy initiatives and decisions. In particular, **1 Policy Recommendation Report** will be elaborated and delivered on M40 that will be discussed in the **policy event** organised during the final conference. In addition, project results will be published in newsletters and journals targeted to policy such as *Science for Environment Policy, Policy Sciences, etc…*

Target Audience: *Government & Plant health policy makers*

Key Quantified targets: ***1 Policy Recommendation Report*** *discussed in the policy event at the end of the project;* ***5 publications*** *targeted to policy-makers.*

Specific communication activities: Communication activities are a crucial aspect of this project, to ensure that we effectively promote our objectives and outputs, as well as its benefits and impacts to the stakeholders we want to reach. As one of the keys to successfully gaining impact is through our ability to secure the cooperation and participation of our key stakeholders, they will be the target of our communication activities. Moreover, it is relevant that citizens are aware of the project developments, societal benefits and positive impacts since the project is benefiting from tax-payers' contributions. Our communication activities will ensure citizens are informed and feel like they are getting value for money. To maximise the impact of our communication activities and guarantee that they will be effectively intertwined with the dissemination effort, the communication plan, as part of the PDEC, will follow 4 steps:

1. **Identify and list all possible stakeholders** - the target audiences of our dissemination activities will be used as the starting point (in addition to any other stakeholders that might be considered important).
2. **Organize our stakeholders** - the target audience of our communication activities will be analysed and categorised into "communities of support" for helping PataFEST to realize its ambition, goals, objectives and impacts.
3. **Development of the PataFEST brand** - in order to help us in engaging with our stakeholders the principles behind the PataFEST brand (visually identified through its logo) will be further elaborated and documented.
4. **Design and implementation of effective communication campaigns (and supporting tools and materials):** We will ensure regular communication to our stakeholders through these campaigns, which will involve the use of emails, face-to-face meetings (for policy-makers), leaflets, posters, the website and social media, etc. A press and media relations campaign will be implemented throughout the project to ensure media interest and coverage.

Clustering and Networking activities: PataFEST will establish **close collaboration and synergies with related projects, networks, clusters and initiatives** at the European and national/regional level to share information and maximise the visibility of the project results. The project will potentially exchange results and deliverables, organize

possible joint dissemination events (**2 intermediate workshops**) and clustering activities with other European projects funded under the same topic or other related topics/projects. In addition, PATAFEST will seek links with the broader dissemination strategies of *the European Foundation for Plant Pathology, the European Centre for Disease Prevention and Control, the European Chemicals Agency, the European Environment Agency and the European Commission's Joint Research Centre, etc.*, at pan-European level, as well as those of National federations and associations of relevance.

A prominent feature to heighten the project visibility is the ***Multi-Actor Participatory Platform*** **(MAPP, T5.5).** It will provide an extended network of relevant stakeholders of the potato value chain across Europe and internationally. The target stakeholders are: **farmers**, **vegetable producers and traders**, **bio-based industry of plant health and soil solutions**, **scientific community**, **consumers**, **policy makers** and **general public**. MAPP will organize, structure and offer a point of information on the topic of the project for supporting users in their valorisation decisions, and, meanwhile, following the open science principles, collect feedback and expectation of target stakeholders & end-users on the project activities and results. The MAPP will also include a special feature with all the citizen of science activities with a link to the *eu-citizen.science* thanks that our partner ECSA is the legacy organisation of this platform. In this way, the planned communication activities will be used best to maximize future dissemination and exploitation of the project results.

Target Audience: *Relevant European and International Organisation; Brother Projects Partners, Potential industrial end-users; Industrial Multipliers and Associations*

Key Quantified targets: ***M48 Final Conference*** *= ≥120 participants;* ***2 Intermediate Workshops =*** *≥60 participants in both;* ***Multi-Actor Participatory Platform*** *= 1;* ***Involvement in network/clusters*** *=project entrance in at least 1 cluster will be evaluated;* ***Inclusion in EU dissemination resources*** *such as the EU Horizon Magazine*

Citizen Science and other activities to engage the general public: PataFEST will promote citizens engagement through **Citizen Science** activities that ECSA has planned in WP 1, WP 4 and WP 5: i) **2 events** (one in the **Long Night of Science in Berlin** and other one in the **Experimental Field Station of ANEC (Spain)** with the purpose of collecting scientific data of potato resistance gens (T1.6); ii) **3 practical workshops** (Germany, Spain and Ecuador) where citizens and owners of small horticulture crops will have the oppotunity to test the bio-based sustainable solutions applied to potato soil (T4.3); and iii) **8 webinars** to promote the participation of citizens in scientific discussion groups with PataFEST' scientist and to disseminate the main technological solutions to tackle potato emerging threats. In addition, the project will develop a comprehensive strategy for activities with the general public, including, press releases, communication campaigns and publication of articles in newspapers and popular magazines. Television is considered a valuable channel in this sense: local channels will be contacted. "**Futuris**" (**"Euronews"** channel) and the **Horizon Results Platform TV** will be considered for a wider coverage. All ther partners, but specially KNEIA, FUND (as coordinator), ECSA (expert in citizen science) and our key agriculture and trade partners (UDAPA, EUPAT and ANEC), will exploit their communication units specialised in maximising media and public outreach as well as building targeted dissemination tools for research and innovation results.

Target Audience: *General public, owners of small crop farms, and mass media (Futuris in Euronews & HRP TV),*

Key Quantified targets: ***Molecular biology experiments with citizens*** *= 2;* ***Co-creative workshops targeting citizens with urban gardens and/or small size crop farms:*** *= 3 (Germany, Spain & Ecuaador);* ***Citizen Science Webinars*** *= 8*

Exploitation Plan and Business Model: A preliminary Exploitation Plan has been drawn up considering the contribution of the results to patents, new or existing technologies, services, infrastructures, knowledge bases and standards, to ensure the project innovations are delivered to the market. Our intention is to realise all the exploitation pathways for the results: *A)* ***Commercial exploitation-*** a key exploitation route will be the industrial use of the project knowledge and technical/technology developments, for the **creation of new technological options, new processes and advisory services to deliver high competitiveness and growth impacts**. *B)* ***Further research-*** providing the knowledge and skills for future spin-out research and development projects will have a high impact for raising the European knowledge base while promoting the targeted industrial sectors' transition to the green economy; *C)* ***Contribution to standards and policies-*** the project results will contribute towards a regulation and standard development and policy revision in the project field. Cooperation with regulatory bodies and third-party sales and distribution licensees in the project field will be implemented. This process has already begun through the concept development and pre-project market validation work carried out by the partners within their existing customer bases. **Key Exploitable Results** of PataFEST will be defined as knowledge, products, technologies and results having a potential for industrial or commercial application in research activities or for developing, creating or marketing the products and processes obtained. Some of the identified results to be exploited are presented below in Table 2.2:

Table 2.2. PataFEST Key exploitable results (KERs)

KERs	Proposed IPR tool	Partners	Potential end users and time to market (after projects end)
Pest monitoring and resistance potato varieties selection	Developing an user friendly bioinformatics tool (*Patview-Patblast*) with all integrated project results, integrated genomic project resources and other resources (Literature, Links, Support Documents), as well as useful tools for performing local and external BLAST searches and other utilities. All resources can be modified/ extended by the user according to the needs. This tool will be of public access and easy to use.	**NEIK** in collaboration with INIAP and UPOL	Scientific community (Universities, RTOs) specialised genetic resources for plant protection. *Time to market: ~2 years*
Phenotype-based High-throughput Screening (**PHTS**) technology	Further internal research; Collaborative Research - results can be used as background of future collaborative research projects.	**UPOL**	Scientific community specialised development of digital tools for pest and pathogen early detection. *Time to market: ~2 years*
VOCs sensors	Patent and license for commercial use; the results will be used as basis for internal and collaborative projects.	**FUND,** IRIS, CRUN	Agrochemical developers, farmers, field machinery & ag-tech producers, advisors, RTOs *Time to market: ~4 years*
AI Prediction Model	Internal service creation to provide AI models to the smart agricultural sector. Collaborative Research	**FUND**	Providers of smart agriculture products & services, Farmers. Agricultural advisers *Time to market: ~3 years*
Mobile hardware for early detection	A *freeium* business model access: i) Integration into existing databases to ensure further scaling of the project results ; ii) PataFEST app containing project outputs (open access) ; iii) Paid services for commercially exploitable results will be developed	**IRIS**	Public and private scientific and technical community. Educational institutions. *Time to market: ~4 years*
Postharvest biocontrol solutions (biocoatings)	Collaborative Research both in EU and National schemes; B2B relationships; internal product development and service creation.	**FUND**, CHEM (formulation scale-up)	Bio-based coating companies. Scientific community. Agriculture sector. Farmers. *Time to market: ~6 years*
Plant protective wax solution	The developed generic plant and crop protective wax solutions can be tailored in future projects for additional specific pests via collaborative future research with public research organisations and private sector; as well for commercial licensing.	TECHNCHEM (formulation scale-up)	
Plant soil solutions (biofertilizers)	Further internal research in collaboration with other research organisations and chemical industries to produce a more generic plant protective wax product.	**UNIBO**, CHEM (formulation scale-up)	

Exploitable foreground will be the basis for the preparation of the **Exploitation Plan** for the companies of the consortium in order to encourage the future investments for market entry of the results of the project. The Exploitation Plan, included as a part of the PDEC, will include the following parts:

1. **Key Exploitable Results** and areas of application, its dissemination on the basis of the communication strategy
2. Analysis of current and target markets:
   - **Market analysis** with a focus on European market, but considering global trends and needs, for all involved

industrial sectors (size, trends, share the technology/product could reach, main competitors, competitive advantage, etc.). It also includes a broad analysis of the stakeholders along the value chain (**stakeholder mapping**), which is relevant for the fine-tuning of the target audience for the specific communication of the results towards the market uptake, to be considered in the exploitation plan. A PEST analysis will be performed to characterize the Political, Economic, Socio-Cultural, and Technological factors and changes to understand the general environment. Markets continuous monitoring will be applied to detect new trends and possibilities and allow the consortium to react and adapt the projects' outcomes to the market changes.

- **Competitive analysis**: identification of the main competitors, (direct and indirect), -to understand the target customers' needs and expectations, and comparison with other solutions in the market or under development. This will be accomplished by performing a Porter's analysis for PataFEST innovative exploitable results and target markets to determine the level of competition expected and support business model development.
- **SWOT analysis** (strengths, weaknesses, opportunities, and threats) to identify the internal and external factors that are favourable and unfavourable for commercial exploitation.

3. **Business model** including marketing strategy and main clients: will be elaborated using CANVAS methodology adapted to the relevant target markets. Value finance will be studied beyond the business model definition and will be approached by assessing the sustainability of business, by identifying future steps strengthening or threatening the business case, by assessing the commercial potential of the technologies, etc.
4. The partners will define IP within the project by relevant fields, results and protection, and the PDEC will cover **IP ownership** with clear definitions of foreground and background information, setting a framework by which all the consortium members can contribute to the exploitation and commercialisation of the project results.

Target Audience: *Government & Plant health policy makers, Farmers, Industry along the potato value chain, Bio-based industry of plant health solutions, Environmental Consulting & Engineering, Innovation programs;*
Key Quantified targets: ***1 Policy Recommendation Report*** *discussed in 2 policy events;* ***Patents*** *= ≥2;* ***Horizon Impact Award*** *= ≥1 application;* ***Exploitation Plan*** *= 1 (Final version delivered on M48);* ***Research Ownership List (ROL)*** *= 1;* ***Joint Ownership Agreement*** *= ≥1;* ***Exploitation Agreements*** *= ≥1;* ***Horizon Results Platform*** *= ≥3 results presented during the project's lifetime;* ***Innovation Radar*** *= ≥3 key project results published;*

Strategy for the management and protection of knowledge and intellectual property rights: The consortium will implement an **integral strategy** for the effective **management of the intellectual property generated in the project, including protection measures.** Key pillars of the IPR management and protection strategy will be: 1) Deep evaluation of the "**Freedom to Operate**", in relation to the background owned by consortium members or potential third parties: the freedom to operate will be constantly assessed in relation to possible patents that might be published in the course of the project; 2) Selection of the IP Manager (KNEIA), as IP expert, to guide the consortium toward the implementation of IPR strategy; 3) The different exploitation routes envisaged will be always aligned and commensurate to the expected impacts (**Destination**), the expected outcomes (**Topic**), the **described impact pathways** and **the project results**, to ensure their viability and opportunity; 4) The strategy will guarantee the dissemination and publication of the results do not interfere with IPR management but serve it: to protect the exploitation potential, a confidentiality policy will underpin our dissemination activities. The project results will be assessed in terms of **novelty, patentability and protection** as part of an effective knowledge management strategy. For the results obtained, ownership, protection and use will be considered in detail. Ownership of patents will be clearly defined and governed by the Consortium Agreement. Results shall be owned by the participant generating them. To avoid doubt or confusion over ownership, a protocol of procedures will be defined, which will be followed by each partner to clearly show the development of the generation of knowledge or results. Whenever the results are generated jointly, they will be **jointly owned**. Allocation and terms of exercising this joint ownership will be agreed in **Joint Ownership Agreements**. However, an overriding principle will be that the **participating industry partners are always provided with the access to the results to exploit them in accordance with the needs and strategy of their business.** Access rights to results for implementing the project will be granted on a royalty-free basis. Access rights for use purposes will be granted either **royalty-free**, or on **fair and reasonable conditions** (established in **Exploitation Agreements** to be negotiated between the requester and granter of the access rights).

Follow-up Plan: The PDEC will include a follow-up plan to foster the exploitation and uptake of the results after project end. The consortium will pay an effort in starting already exploitation actions during the project execution. Nevertheless, if no exploitation takes place, following Horizon Europe indications the consortium will use the Horizon Results Platform for making their exploitable results visible, during the first year after the end of the project. In addition, the MAPP, including the citizen of science activities with a link to the *eu-citizen.science* will be active at least 9 years after the projects end (KNEIA will be responsible).

2.3 Summary

Specific Needs	Expected Results (linked to project's results)	D & E & C measures	Target Groups
In 2020, the EU's potato production was 55.3 million tonnes (=€12.3 billion). Globalisation and climate change are the main drivers for entrance of emerging or new potato pests to EU: **CLso** is spread by the vector psyllid **BC** causing **Zebra Chip** with severe potato yield losses (up to 100%). Excessive use of pesticides and the transition towards their bio-based counterparts, increase **resistance** in bacteria, insects, and soil-borne pathogens, and reappearance **of soil pathogens**, potentially increasing the incidence of soil infections during postharvest storage. To preserve the health of potato plant and the quality of the final product against its emerging pests, the project PataFEST studies different solutions for early detection and prevention and incorporated in a **IPM strategy.**	PataFEST expects to generate **18 results (R)**: **Early detection:** - Monitoring and control protocol of CLso spreading and its vector BC (**R1**), mapping of potato resistance genes (**R2**) and marker-assisted selection in molecular potato breeding (**R4**). - Analysis of 500 samples for postharvest disease resistance and 150 of Ecuadorian samples for CLso and its vector BC resistance (**R3**). - Identification of biomarkers in infected plants and tubers to be used in the development of an image analysis digital tool (mobile app) and VOCs sensor using improved with AI-based models (**R5-R6**). **Prevention strategies:** - Plant solutions against CLso (green wax sprays), natural soil management strategies (biofertilizers) and biocontrol coatings against soil borne pathogens (**R7-R9**) **Validation of the pest management solutions** in field trials at selected EU locations and Ecuador (**R10-R13**) **Citizen Science strategies** to engage general public and farmers and to collect relevant scientific data (**R14-R15**). **Multi-actor approach** to share knowledge, experience and ensure economic sustainability of EU potato production through an effective dissemination & communications plan and creation of "practice abstracts" to the EIP-Agri repository **(R16-R18)**	6/8 Scientific publications. 60 EIP-AGRI practice abstracts. 24 Conferences and events. 1 Horizon Impact Award. 1 Conference proceeding per year per partner. 2 Intermediate workshop. 1 Final conference. Logo, website, videos (x2), press releases, social media, infographics, roll-ups, leaflets, posters. 1 Policy Recommendation Report 5 publications to policy-makers. Multi-actor participatory platform + Citizen of Science events (2 scientific events, 3 experimental field workshops, 8 webinars. 2 Patents, Exploitation Plan, Research Ownership List (1), Join Ownership Agreement (1), Innovation Radar (3)	Scientific/Technical community Farmers Industry along the potato value chain: Breeders Producers Traders Associations and Co-operatives Bio-based industry of plant health solutions Environmental Consulting & Engineering Citizens Plant health policy makers Sandarisation bodies National authorities European Commission

OUTCOMES	IMPACTS (linked to Expected Impacts (EI) of the destination and KIPs)
OC1: Relevant and free access information to better identify the most suitable potato varieties resistant to plant pest threat and soil-borne pathogens in postharvest storage. **OC2**: Early detection tools and sustainable solutions to improve plant and soil health and to control postharvest pathogens aiming the reduction of fossil-based pesticides and fungicides applied in potato crops. **OC3**: Field trials both, in potato plant and in tubers, during postharvest storage of the technologies developed at TRL5 to validate the effectiveness of the IPM strategy of potato emerging threats and disease. **OC4**: Use of citizen science activities in the agriculture sector as an effective supplement to pest management approaches. **OC5**: Provide the foundations and drivers for future development and implementation of IPM strategies by farmers and other key players in the event that these emerging potato pests outbreak in EU.	- Of the 500 analysed samples, at least 5% potato resistance varieties for targeted diseases and pests will reach the market (**EI 3 → KIP 1**) → Replicability of PataFEST molecular characterisation strategies to other similar crops (e.g. carrots, pepper, etc.). - Development of sustainable innovative solutions (including cutting edge digital techs) for early detection, surveillance, treatment and (bio)control of potato pests in pre-harvest and post-harvest strategies: 50% reduction of BC, 25% reduction of ZC, 40-50% reduction of soil-borne pathogens, 90% of reliability in the detection of CLso-infected potato (**EI 1 → KIP 7 & 9**) → Further development to go from TRL 5 to TRL 7 to achieve an Integrated Pest Management to protect EU potato production. - Scientific data is also produced by citizens using citizen science initiatives and activities that will be active beyond the project's duration (**EI 3 → KIP 6**) → the platform eu-citizen-science will hold a special future of the project's citizen science activities along its duration and beyond (~9 years). - Creation of an interactive innovation model to help the implementation of potato IPM across different professional backgrounds ('Multi-Actor Approach') (**EI 1 & 3 → KIP 5 & 5**) → The multi-actor participatory platform will be active at least 9 years after projects ends.

3 IMPLEMENTATION

3.1 Work plan and resources

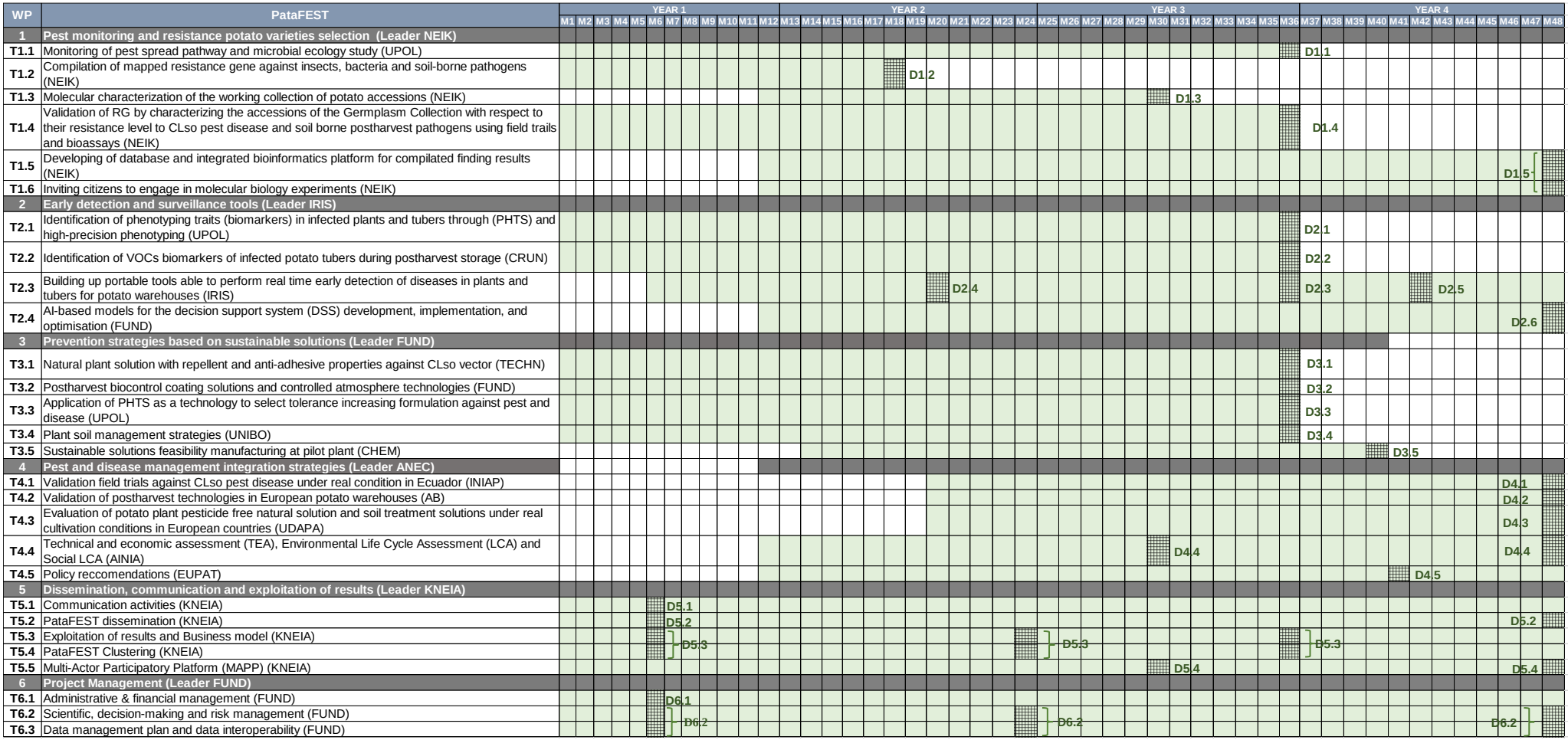

Figure 3.1 PataFEST Gantt Chart.

PataFEST work plan will be implemented in 48 months and it is divided into 6 work packages aimed to fulfill the general requirements set out by the target HE 's topic and promoting the interaction between all participants.
In the first part of PataFEST, **pest spread pathways** and the **main resistance potato varieties** against CLso and postharvest pathogen will be explored at molecular level (WP1). Then, PataFEST will develop early detection and surveillance tools as **image analysis digital tool (mobile app)** and **portable VOCs sensor** for the rapid diagnostic of the pest and pathogens diseases development (WP2). Besides, PataFEST will include the development of pest prevention strategies based on sustainable and effective **plant** and **soil treatments** and **biocontrol coating solution** to control incidence of soil-borne pathogens during the potato postharvest storage (WP3).

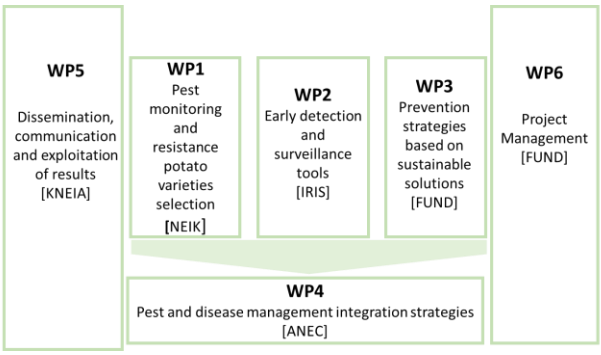

Figure 3.2. PataFEST Work Plan

In a second part of PataFEST, environmentally friendly and innovative mitigation strategies developed will be integrated and validated in a relevant environment (TRL5). Dedicated dissemination, communication and exploitation activities will secure the maximum impact of the project (WP5). Finally, WP6 deals with the management of the project in order to make all the proposed developments in line to the PataFEST objectives. Figure 3.1 provides an overview of the work packages and their timing (Gantt Chart) and Figure 3.2 shows PataFEST work plan structure.

Work Package No.	**WP1**	Starting date / Ending date	M1 / M48
Work Package title	**Pest monitoring and resistance potato varieties selection (Leader: NEIK)**		

Objectives

- Development of the BC and plant sampling protocol.
- Optimisation of the specific primers for CLso and BC detection.
- CLso spreading pattern analyses across the target area.
- Development of monitoring protocol for surveillance, detection, and control of plant pests and its vector.
- Set up a catalogue of mapped RGs against insects, bacteria, and soil-borne pathogens.
- Molecular characterisation of the working collection of potato accessions.
- Validation of RGs in the working collection of potato accessions with respect to CLso pest disease and soil-borne postharvest pathogens.
- Preparation of the obtained Results for Publication and Dissemination.

Description of Work

T1.1. Monitoring of pest spread pathway and microbial ecology study (M1-M36). Leader: UPOL. Participants: INIAP.

Subtask T1.1.1 Sample design of the monitoring pipeline. *Leader INIAP*

To monitoring the vector BC, yellow sticky traps will be used in the field to count adults of the potato psyllid and direct observation for the other development stages. To capture the specimens, suction vacuums or net will be used. The insects will be preserved in alcohol and later the entire adult will be used for the molecular analyses. Complete plants will be taken from the field and transplanted into pots for their movement from the field to the laboratory. Several parts of the plant will be used such us petiols, aereal tubers, stolons, and underground tubers for DNA extractions. Plant tissue samples will be taken from different plant development stages, including tuber when they form.

Subtask T1.1.2 Molecular characterisation of ecological pest spread pathway. *Leader UPOL. Participants: INIAP*

To identify genetic diversity and spread patterns UPOL will use the result of the standardised collecting protocol of the pest samples from the T1.1.1. The DNA will be extracted using the available commercial kits. A modelling approach to surveillance, detection, and control of CLso pest and its vector will be carried out using different molecular DNA base approaches. Initially, UPOL will review and collect all public genomic information of BC and CLso as assembled genomes, 16S barcodes and commonly primers used for the target taxa detection. Then, in order to extend the existing molecular knowledge with the specific haplotypes that are present in the working area, the whole genome of the CLso sampled across Ecuador will be sequenced with the aim to produce 150-200 high quality bacterial genomes using PacBio Sequel II HiFi reads. The genomes will be assembled using SMRT Link bacterial assembler and the genes will be annotated by the RAST web service. Moreover, GWAS analysis will be performed among all the sequenced genomes to identify the molecular diversity and potential functional differences that can connect with the pathogenies of the CLso pest and disease response. The vector population will be monitored for the CLSo strain type pooling several individuals per location. The target gene will be 16S rRNA sequences, a commonly extended marker used for classification and identification of bacteria and archaea. 16S rRNA genes of distinct regions will be amplified and taxonomically identified using publicly available NCBI GenBank and Q-bank databases (500-

1000 samples). Combining precise location information and the collecting data of the samples, UPOL will infer spreading patterns through the target area using a continuous phylogeographic approach. Moreover, to optimize the accuracy of qPCR detection, UPOL and INIAP will critically analyse the previous findings and design the set of unique primers to specific target pest strains and its vector. Then, a surveillance monitoring protocol of CLso strains and BC will performed using optimised qPCR approach (1500-2000 qPCR assays).

T1.2. Compilation of mapped resistance gene against insects, bacteria and soil-borne pathogens (M1-M18). Leader: NEIK. Participants: INIAP

To carry out this task, the following resources will be exploited, in order to get the targets that modulate the defence against the pathogens: **i)** annotation and compilation of already mapped RGs on the SNP GGP V4 chip (35K) Potato array, (**ii)** annotation, compilation and mapping to the Potato Reference Genome of Potato RGs available at NCBI and from the Potato Genome Sequencing Consortium (PGSC) at SPUD DB; **(iii)** compilation of relevant RGs from the Plant Resistance Gene Database (PRGdb) and identification of Potato orthologues and mapping of these genes to the Reference Genome.

Finally, considering the targeted sequencing task (T1.3), in addition, an inventory of available SNPs within the genes at NCBI (dbSNP), SPUD DB Genome browser (Track GBS SNPs) will be established.

T1.3 Molecular characterisation of the working collection of potato accessions (M12-M30) Leader: NEIK. Participants: INIAP.

For this purpose a broad germplasm working collection will be analysed based on: **i)** Germplasm collection from NEIKER with 40 wild species, 35 native potato varieties and 230 old and new European cultivars; **ii)** around 150 samples (DNA aliquots) from INIAP which will be characterised for their tolerance level to both, CLso and the insect vector and; **iii)** potato cultivars and accessions of interest from other Project partners (DNA aliquots and tuber materials for validation will be provided).

A total of up to 500 accessions describe above will be genotyped by the following methods: **i)** characterisation of all germplasm collection accessions using the SNP GGP V4 chip (35K) Potato array manufactured by NEOGEN Europe; **ii)** integrated SNP and targeted sequencing through SPET technologies using a selection of additional important R-genes <u>not included</u> on the previous chip in all germplasm accessions.

T1.4 Validation of RG by characterizing the accessions of the Germplasm Collection with respect to their resistance level to CLso pest disease and soil-borne postharvest pathogens using field trails and bioassays. (M1-M36). Leader: NEIK. Participants: INIAP, UPOL.

Subtask T1.4.1 Selection and multiplication of plant material for bioassays. *Leader:NEIK.*

NEIK will carry out the multiplication of selected plant materials for the bioassays in T1.4.3 and for other partners involved in WP1(T1.4.2) and WP2 to include these materials in their planned experiments. The required in vitro plants and minitubers of local and introduced germplasm will be produced using tissue culture, stem cuttings and hydroponic or aeroponics technologies in automated greenhouses. In addition, INIAP will perform PCR tests for CLso during the process to ensure the production of free CLso tuber seed. At least 20 in vitro plantlets and 10 kg of seed tubers per genotype will be produced for using them in the proposed activities.

Subtask T1.4.2 Characterisation of the tolerance against CLso and potato psyllid vector. *Leader: INIAP.*

The characterisation of tolerance level to both, CLso and the insect vector will be performed from around 150 samples supplied by INIAP consisting of 100 samples of advanced breeding materials, wild species, native species and at least 50 European potato varieties using greenhouse and field trails. BC adults obtained from a CLso-positive INIAP´s colony will be reared on potato plants. Adult behaviour, mortality, and number of adults, eggs and nymphs will be measured. Symptoms description in plant and tubers will be recorded. In field conditions under natural BC infestation symptoms, severity and incidence of zebra chips in plants and tubers, insect mortality and presence of eggs will be recorded. At harvest time, yield and fried chip/tuber quality (score from 1 absence -5 extreme symptoms) will be evaluated. The PCR test will be used to confirm the presence of CLso in the different experiments.

Subtask T1.4.3 Characterisation of resistance level to postharvest diseases. *Leader: NEIK.*

Other accessions of the Germplasm Collection will be characterised with respect to their resistance level to postharvest diseases (dry rot caused by *F. sambucinum*, black dot caused by *C. coccodes*, and silver scurf provoked by *H. solani*) using bioassays. For Fusarium susceptibility evaluations potato mini tubers will be wounded internally and inoculated with agar plugs containing the pathogen. After 20 days the depth and width of the rot area are measured. For Colletotrichum and Helminthosporium susceptibility evaluations potato mini tubers will be planted in artificially infested soil with conidia (H) or microsclerotia (C), respectively. For disease assessment the tubers will be examined after incubation for the presence of setose microsclerotia of *C. coccodes* or the characteristic conidiophores of *H. solani*. The number of lesions will be counted and/or the percentage of the surface area covered by sporulating lesions. Complementary, an automated rapid PHTS using RGB-imaging in controlled conditions developed in WP2 by UPOL will be employed to increase the testing throughput. In a phenotyping device condition of a commercial warehouse will be simulated to cover broader spectrum of postharvest scenarios.

Subtask T1.4.4. Genome-wide association studies (GWAS). *Leader NEIK.*

Association studies will be performed using GWASpol (GWAS using different models) between observed phenotypic resistance level data and molecular SNP data from WP1. In addition, the predicted performance data from WP2/WP4

experiments will be also used as phenotypic data for association studies and the results will be compared. Finally, a selection of promising RG for each targeted disease or pest based on the association signals results.

T1.5 Developing of database and integrated bioinformatics platform for compilated finding results (M12-M48). Leader: NEIK. Participants: INIAP, UPOL, ECSA.

Firstly, a compilation of all results obtained in WP1 in form of databases and derived publications will be performed. Then, a user-friendly bioinformatics tool (PATVIEW-PATBLAST) will be developed with all the integrated project results, integrated genomic project resources and other resources (Literature, Links, Support Documents), as well as useful tools for performing local and external BLAST searches and other utilities. All resources can be modified/ extended by the user according to the needs. This tool will be of public access, easy to use and developed by NEIK.

T1.6 Inviting citizens to engage in molecular biology experiments (M12-M48). Leader: ECSA. Participants: FUND, NEIK, UPOL, ANEC.

ECSA and ANEC will coordinate 2 molecular biology experiments involving citizens of all backgrounds with a strong interest in science to promote scientific discussion with PataFEST scientists from FUND, NEIK and UPOL and to generate scientific data of potato RGs. One workshop will be hold in the "Long Night of Science in Berlin", and the other one in the "The Doctor's Farm, Experimental Field Station" of ANEC. The experiments will consist of DNA-extraction and PCR-sequencing of potato samples supplied by the participants. For both events, main results obtained as information about potato RGs, molecular biology methods for potato health and project updates will be disseminated in the citizen science webinars (WP5).

Deliverables: D1.1 Monitoring results of CLso spreading pattern (UPOL, M36); **D1.2** Catalogue of Potato RGs with SNP information (NEIK, M18); **D1.3** Genotypic Potato Chip data and SPET data (NEIK, M30); **D1.4** Association study results, predicted performance data and selection of most promising RG for pest and postharvest diseases. (NEIK, M36); **D1.5** Result databases and bioinformatics tool for publication and dissemination actions (NEIK, M48).

Work Package No.	**WP2**	Starting date / Ending date	M1 / M48
Work Package title	**Early detection and surveillance tools (Leader: IRIS)**		

Objectives

- Develop image analysis (e.g., smartphone) application for the early detection and surveillance of the plant pest diseases.
- Develop low-cost diagnostic solutions as VOCs sensor devices to detect postharvest pathogens in potato warehouses.
- Define key features of for the future and emerging pests and disease spreading (climate changes and pest globalisation).

Description of Work

T.2.1 Identification of phenotyping traits (biomarkers) in infected plants and tubers through (PHTS) and high-precision phenotyping (M1-M36). Leader: UPOL. Participants: CRUN.

Subtask T.2.1.1 Rapid PHTS for validation of the testing pipeline through proof-of-concept with model potato genotypes and pathogens. *Leader: UPOL*

Rapid PHTS method will be adopted by UPOL for potato plants and microtubers of model potato accessions (with known contrasting level of pathogen resistance) and pathogen species defined by WP1. Segments of *in vitro* generated potato shoots will be grown in multiwell plates in HTS-vertical phenotyper device with controlled conditions. Top-view RGB imaging will be performed during the cultivation to obtain series of RGB images documenting time-dependent progression of morphological and colour changes in potato shoots infected with selected pathogens. In the case of microtuber phenotyping, conditions of a commercial warehouse will be simulated to cover broader spectrum of postharvest scenarios. Libraries of RGB pictures obtained will be used as training datasets for development of the AI-based applications for the early detection and surveillance of the plant pest diseases in T.2.4 and their validation.

Subtask T2.1.2 High-precision phenotyping of potato plants and tubers for detection of biomarkers and characterisation of the accessions selected for potential resistance. *Leader: UPOL.*

High-precision phenotyping at OloPhen of UPOL using non-invasive imaging with different sensors (RGB, IR, hyperspec, Fluorcam) to analyse the development of the changes caused by selected pests on potato plants and pathogens on tubers in controlled conditions will be developed. Multivariate statistical approaches will be used to identify both morphological and physiological traits as biomarkers of the infection and the most informative sensor(s) for their detection. The identified biomarkers will be used in T2.3 and final protocol for characterisation of the accessions of the Germplasm Collection selected in T.1.3. This protocol will be also used for analysing the effects of natural and sustainable solutions for potato plant and postharvest tubers in WP3 for the identification of concentration ranges that are not negatively influencing the plant growth and development.

Subtask T2.1.3 Phenotyping potato plants and tubers for the validation of the biomarkers identified in T2.1.2. *Leader: CRUN.*

Infected and control commercial cultivars and accessions selected for their resistance to infection will be grown under controlled glasshouse conditions. Above ground images will be taken at least at five stages throughout plant development for early detection of disease development (T3.4). Harvested tubers will then be stored under commercial

storage conditions and further phenotyped using RGB and hyperspectral cameras at regular intervals (e.g., monthly); multivariate and machine learning approaches will be used for tuber classification based on disease evolution.

T2.2 Identification of VOCs biomarkers of infected potato tubers during postharvest storage (M1-M36). Leader: CRUN. Participants: FUND.

Subtask T2.2.1 Identification of VOCs and potential biomarkers related to the infection of potato tubers with the selected pathogens. *Leader: CRUN.*

CRUN will analyse the headspace of small potato microcosm stored under conducive conditions for disease development using Thermal Desorption-Gas Chromatography-Time of Flight-Mass Spectrometry (TD-GC-ToF-MS). Biomonitoring TD tubes will be used for the selection of low/medium molecular weight compounds. The methodology has been selected as allows a high fold concentration of the volatile compounds produced. An initial identification of compounds and a non-targeted metabolomic analysis will be carried out using specific format files (ELU and FIN files) by AMDIS (Automated Mass Spectral Deconvolution and Identification System) based on the 2014 NIST library[56]

Subtask T2.2.2 Quantification of VOC biomarkers under simulated postharvest storage conditions. *Leader. CRUN.*

The preliminary target compounds identified in T2.2.1 will be investigated by CRUN. Both, commercial potato cultivars (n=2) and varieties selected from WP1 will be stored under simulated commercial scenarios. Once presence of some key VOC species has been confirmed the right standards will be purchased from VOCs and calibration curves built for analysis determination of VOC concentrations.

T2.3 Building up portable tools able to perform real time early detection of diseases in plants and tubers for potato warehouses (M6-M48). Leader IRIS. Participants: FUND, CRUN, UPOL.

Subtask T2.3.1 Development of a portable VOCs sensor for potato warehouses. *Leader: FUND.*

FUND will develop an array of electrochemical sensors for the early detection of potato diseases. Electrochemical sensors will detect the concentration of VOCs selected as biomarkers in task 2.2. For this purpose, FUND will select suitable supports for the sensitive material, focusing mainly on Interdigital electrodes and three electrodes configuration. Sensitive material will be based on MOFs or conductive polymers such as Polyaniline or Polythiophenes compounds which are able to detect several VOCs. Suitable electrochemical technique will be employed (potentiometry, amperometry or chemiresistive) depending on the sensing material selected for the optimal biomarker. Different inks containing sensitive materials, binders, surfactants, and other additives will be developed in to cover the surface of the electrodes. Each sensor will be characterised using electrochemical techniques and microscopy. The array of sensors will be tested with pure, binary, and tertiary mixtures of biomarkers in order to know the possible interferences among sensors. In case of the interferences were high, an e-nose could be developed in order to improve the selectivity and accuracy of the array to make predictive measures. IRIS will integrate the sensors developed by FUND into a printed circuit board (PCB), this PCB will contain components for powering up the sensors and signal conditioning components for measuring changes in the range of mV and µA. The PCB will be able to convert the electrical signal coming from the sensor array, generating a digital signal compatible with the user platform. Finally, a full sensor prototype (array plus electronics) will be validated using accelerated tests at lab scale introducing the sensors in a chamber containing 10Kg of infected potatoes.

Subtask T2.3.2 Digital platform design: Specifications, methods and preliminary requirements. *Leader: IRIS.*

The initial specifications and protocols for the digital platform design will be determined. In collaboration with other partners involved in WP2, IRIS will perform an exhaustive study for the road mapping and designing of the different layers and domains that will conform the digital platform. Also, an assessment of the data sources and data formats will be performed to upgrade the government, utilisation, and optimisation of the resulting platform.

Subtask T2.3.3 Mobile app development, integration, and optimisation. *Leader: IRIS.*

IRIS will develop the digital platform for the data gathering and data analysis; based on the specifications resulting from T2.3.2. Here, a compendium of micro-architectures supporting a global backend-frontend system will be produced. The digital tool will count also with an adaptable application allowing it to operate also from portable devices such as tablets and smartphones. The system will also include a user-friendly top layer for practical and satisfactory experience, optimised to the end-users needs. In addition, the platform will be able to host optimised AI-based algorithms (Decision Support System - DSS), which will be the core element for the real-time in-field image analysis of the potato contaminants with a single shooting of the product. After the system has been optimised in the next task 2.4, it will be ready for deployment and validation (WP4).

T2.4: AI-based models for the decision support system (DSS) development, implementation, and optimisation (M12-M48). Leader: FUND. Participants: IRIS, ECSA.

The DSS will be developed in accordance with the insights determined and provided by tasks T2.1 and T2.2 regarding the biomarkers and VOCs detection and identification. Employing the raw data provided from lab/pilot scale trial (T2.2) and commercial storage (potato warehouse trials in WP4) will be processed by Machine Learning techniques to provide a higher-level information, and fused with different data sources, related to the KPIs. The DSS will include two very differentiated sets of AI-based algorithms: one set will be correspondent to the image-analysis for the real-

[56] National Institute of Standard and Technology, US Government.

time detection of potato pest diseases; and another set will be linked to the prediction of pest and disease spreading based on climate change and globalisation influences.

Subtask T2.4.1 Context risk model of pest and disease spreading based on climate change and globalisation influences (Data driven climate model). *Leader: FUND.*

The objective is to identify whether current meteorological and environmental conditions are favourable for the CLso spread in European territory, in accordance with the scenario already existing in Ecuador. Furthermore, it will be studied whether climate change will generate these conditions in Europe and an estimate will be made regarding the timing of its appearance in case the data can represent this knowledge. For this purpose, there will be developed a model capable of comparing the context and ecosystem of pest-free areas, particularly Europe, and areas where the pest has been detected in Ecuador. The model will be powered by current and historical climatological data provided by ECSA, as well as other possible data on the current and past context taken from bibliographic sources, focusing especially on the bibliography on environmental forecasts caused by climate change.

Subtask T2.4.2 Predictive model development based on AI algorithms for the early detection tool (app) of the pest and diseases. *Leader: FUND.*

In this subtask, a predictive model based on Machine Learning techniques will be designed and developed by FUND for the prediction/classification of pest diseases mobile app. The data used to learn will be provided from the study of phenotypes (T2.1), VOCs biomarkers (T2.2), and resistance gene information including GWAS results (WP1). The data in image format drives the techniques used from the Deep Learning area, with all other inputs included in the model as contextual or complementary information. After the model has been designed and implemented, it will be trained in a first stage with the available data. After that, the machine learning models will be validated on WP4, both on plant and postharvest potato tubers. The data generated during the validation process will be used to refine the models and improve the quality of the results.

Deliverables: D2.1 Identification of phenotyping traits in infected plants and tubers (UPOL, M36); **D2.2** VOCs biomarkers identification and quantification results (CRUN, M36); **D2.3** Development of a portable VOCs sensor for potato warehouses (FUND, M36)**; D2.4** Alpha version of PATAFEST digital platform design development for validation (IRIS, M20)**; D2.5** DSS design, development, and optimisation for the early detection of potatoes diseases and pest spreading. (IRIS, M42)**; D2.6** AI-based algorithms developments for the real-time detection of potato diseases and prediction of pest and disease spreading (FUND, M48).

Work Package No.	**WP3**	Starting date / Ending date	M1/ M40
Work Package title	**Prevention Strategies Based on Sustainable Solutions (Leader: FUND)**		

Objectives

- Provide natural and sustainable solutions to prevent pest spread by controlling the vector in potato plants.
- Investigate biocontrol solutions and controlled atmosphere strategies to reduce postharvest disease incidence and severity.
- Reduce the incidence of CLso and post-harvest pathogens in potato crop increasing plant health status thanks to the optimised nutrient availability and improved plant soil management.

Description of Work

T3.1 Natural plant solution with repellent and anti-adhesive properties against CLso vector (M1-M36).

Leader TECHN.

TECHN will study the evaluation of various formulations to assess different possible processes to prepare coatings based on their wetting and self-cleaning properties on a plant leaf model substrate. Concentration screening of specific fatty acids and solvents; air/liquid ratio and rate of spraying will be determined.

In parallel to the previous activity, TECHN will screen essential oils as repellent solution against psyllids BC (via potato aphid mode such as *Myzus persicae)* and will study their integration into the wax precursors. Screening for repellence and deterred oviposition using of the model will employ laboratory bioassays of both choice and non-choice tests. Subsequently, the most potent essential oil mixtures will be incorporated into the wax precursors.

Tests with the most promising wax formulations (passive and with added essential oils) will be carry out to establish relationships between the coating structure/composition and the resulting wetting, thermal and mechanical properties characterisation. Wetting properties for coating quality will be assessed via state-of-the-art characterisation techniques for the texture, crystallinity, preferred orientation, size, and strain of the formed wax crystals size; different microscopy determinations for mechanical, chemical composition and homogeneity; thermal stability analysis.

The results collected will be used as an initial benchmark for further formulations that will be developed.

The stability of the most promising coatings against different types of environmental stresses will be evaluated by exposure to various levels of humidity, UV light and temperature conditions for extended periods of time. First screening will be performed at TECHN. Then, a suitable test schedule will be established with sets of normed test procedures to mimic conditions relevant to coatings for potato plant leaves. These tests will be carried out using specialised climate chambers that will simulate various varying weathering conditions present.

Comprehensive study of the antibiofouling, antimicrobial and vector-repellent properties of the resulting coatings will be carried out. Anti-adhesive properties provided by the wax coatings on target substrates and different environmental conditions will be investigated and quantified.

Characterisation of leaves photosynthetic activity, transpiration rate, and super-hydrophobicity by wax coating treatment will be evaluated. Morphological features of the coatings using different spraying techniques and conditions will be assessed using SEM. Coating thickness, uniformity, its elasticity during leaf tissue expansion and persistence will be evaluated under SEM observations.

T3.2 Postharvest biocontrol coating solutions and controlled atmosphere technologies (M1-M36). Leader: FUND. Participants: CRUN.

FUND supported by CRUN will carry out in vitro and in *in vivo* explorations testing biocontrol effect of both Lactoperoxide system enzyme (LPS) and antagonistic yeast incorporated into edible coatings against soil-borne pathogens responsible of the postharvest potato diseases as *C. coccodes*, *H solani*, *F sambucinum*.

For the in vitro analysis, Lactoperoxidase system enzymes (LPS) will be prepared adjusting the ratio of the different components and the enzymatic activity will be assessed. *W. anomalus* will be also prepared at different concentrations to be tested against target soil-borne pathogens. Turbidity test will be carried out against the soil-borne pathogen candidates to identify the optimal antimicrobial concentration of LPS and antagonistic yeast.

Once obtained the optimal antimicrobial concentration, solution of enzymes and antagonist yeast will be incorporated into water based edible coating. Coating solutions will be composed by several ingredients/additives as polysaccharides, lipids ana other functional compounds (surfactants, emulsifiers, antioxidants, etc.) edible coatings should provide improved barrier to gases and good mechanical properties to completely cover the surface and reduce senescence of the products treated.

In vitro studies will be performed to assess the antimicrobial activity of film-forming solutions including agar disk diffusion, dilution methods, spore germination assay, enumeration by plate count method, and film surface inoculation test. Besides, barrier properties as oxygen and water vapour permeability, mechanical properties, gloss, viscosity, colour, surface tensions etc, will be determined.

Formulation prototypes, that best results showed in in vitro studies, will be tested in *in vivo* on potato surface, and stored under control (air/commercial industrial conditions) and different controlled atmospheres (CO_2:0,5-10%; O_2:2-5%) Real-time respiration rate and pathogen incidence (presence or absence) and severity (% of total surface affected) and quality parameters as starch content, dry weight, reducing sugars will be evaluated during storage. Moreover, a full metabolomic approach will be carried out by means of LC-MS/MS to identify non-volatile biomarkers, such as phenolic compounds, non-structural carbohydrates. and plant growth regulators linked to the development of the pathogens during potato postharvest storage.

T3.3 Application of PHTS as a technology to select tolerance increasing formulation against pest and disease (M1-M36) Leader UPOL. Participants: FUND, TECHN, UNIBO.

Rapid PHTS method with model plant Arabidopsis will be used by UPOL for simultaneous assessment of potential phytotoxic effects of natural plant solution with repellent and anti-adhesive properties developed by TECHN. Individual components of the formulation, their combinations in different concentration ratios will be screened by foliar application on Arabidopsis seedlings grown in 24-well plates in controlled conditions in a dose-dependent manner. Periodically, the dynamics of growth and colour changes in the rosettes will be monitored using non-invasive top-view RGB imaging. The obtained morphological and physiological traits of the treated and non-treated plants will be compared and the components or their combinations with negative effects will be reported. The overall effect of the candidate formulations for the final product on the morphological and physiological status of potato plants will be then assessed by multisensor PHT Potato plants of selected genotypes will be grown in pots in the OloPhen platform under controlled conditions simulating the targeting field environment. The repellent solutions will be sprayed in different concentrations and doses and the plant vigour will be analysed to the detail by RGB, thermal, hyperspectral, and fluorcam sensors. Similar approach will be applied in the case of testing of the postharvest biocontrol coating solutions developed by FUND. Rapid PHTS method will be used to support the quantitative testing of the effect of formulation prototypes on pathogen incidence (presence or absence) and severity (% of total surface affected).

T3.4 Plant soil management strategies (M1-M36). Leader: UNIBO. Participant: CRUN.

Chitin (chitosan) and calcium carbonate-based products, obtained from processing of biomasses residues and agricultural waste (e.g., seafood), will be used as carrier(s) of nutrient elements. Carrier(s) will be bound to phosphate group(s) and functionalised with elements (micronutrients). The efficiency of the functionalisation will be proved by elemental analysis (ICP) and release test in selected chemical media (e.g., relevant buffers). The compounds obtained will be applied, as amendments, directly on soil in experimental potato production plots under controlled conditions. Different ways of application (liquid dispersion, solid etc.), doses and treatment timing will be compared.

The trials will evaluate the treatments effect on soil elements bioavailability of each microelement content in potato leaves and in tuber at different stages of the production cycle. A general nutritional status of the plants will be also checked at the end of each growth cycle of the plants. Both untreated and chemical fertilised soils will be considered as control treatments. The best experimental conditions will be repeated in a second cycle under controlled conditions extending the trial to further potato varieties to improve robustness of results obtained.

To test the effects on the potato plant health status of the innovative fertilisation under CLSo infection, trials will be carried out exclusively in a dedicated section of UNIBO greenhouse following the European standard good practices. Trial experiments will be set on potato plants from the better performing variety, chosen according to the accumulation of selected elements that will be grown on untreated soil, as negative control, and on soil where fertilisation will be applied by previously defined protocols optimizing the nutrient bioavailability.
Under the described experimental conditions, the pathogen infection will be applied at different times, within growing stages from sprout development and tuber initiation, in different and subsequent experiments. Plant infection will be obtained by grafting to decipher the direct effect of nutrients on pathogen development on its host.
The infection level and the biological activity of CLSo in potato tissues (leaves and tubers) will be assessed by visual screening and, respectively, by a Droplet Digital PCR (ddPCR) and RT-ddPCR.
The experimental conditions returning the highest protection level of potato plants against CLSo, will be repeated as previously described in a new complete production cycle, including different potato varieties, to validate the results obtained. Tubers produced under the described experimental conditions will be also assessed for resistance to post-harvest pathogens. Specifically, tubers collected from trials showing improved nutrients elements accumulation will be used in post-harvest conditions experiments to evaluate changes in the susceptibility to post-harvest pathogens such as *F. sambucinum*. *C. coccodes*, and *H. solani*, Before or during storage, artificial inoculation will be applied sparing a conidia suspension of the selected pathogen at wounded surface of the tubers at different time after injury.
According to the pathogen behaviour the progression of the infection will be monitored at different time by visual scoring while data analysis will search for tuber lots from fertilisation trials that minimize pathogen growing when compared to tubers form untreated soil.
At CRUN the effect of extreme climate conditions (e.g., prolonged period of drought) as expected under future climate change scenarios on potato plant growth, tuber production and disease severity will be tested under controlled environment conditions. Large 1 m^3 lysimeters in the new Agri-Tech supported Plant Phenotyping and Soil Health facilities will be used to grow potatoes in a well-draining sandy loam soil. Lysimeters are placed on balances to record water usage of each and steer an automated irrigation system, supported by soil moisture sensors recording wetness at depth, to maintain predetermined soil wetness and allow to impose drought stress. Triplicate lysimeters with and without drought stress and plus or minus introduction of diseases (viz., black dot, silver scurf, dry rot) will be compared for the effect on growth, tuber development, tuber disease severity upon harvesting and following storage in our postharvest facilities. Belowground cameras will be used to monitor tuber and root development; a phenotyping platform will scan all lysimeters in the glasshouse at fortnightly intervals. Sensors include 3D laser (monitoring growth), Thermal (stress, transpiration), hyperspectral (pest and disease) and RGB (development, pest, and disease). Results will be correlated with those observed in the experiments carried out in T2.1.3.
T3.5. Sustainable solutions feasibility manufacturing at pilot plant (M14-M40). Leader CHEM. Participants: FUND, TECHN, UNIBO.
CHEM as an expert industrial developer of crop protection products developed, will be in charge to manufacture natural plant solution with repellent and anti-adhesive properties, postharvest biocontrol coatings and alternative biofertilizer soil treatment at pre-industrial (pilot plant) scale (TRL5). Several optimisation steps will be further included to have a defined production protocol for each final product. FUND TECHN and UNIBO as solutions developer, will support CHEM on the manufacturing process. Periodically, functional activity of each solution manufactured will be evaluated during a defined storage shelf life (maintaining antimicrobial activity, e.g.). Besides, pre-industrial manufacturing will be carried out according to standard of quality, safety, and respect for the environment (toxicity assessments). For this, quality and toxicity analysis will be determined on the solutions produced. Finally, required amounts of each solution will be manufactured to be tested in field validation trials during the WP4 execution.

Deliverables: D3.1 Natural plant solution with repellent and anti-adhesive properties developments (TECHN, M36); **D3.2** Postharvest biocontrol coating solutions and controlled atmosphere technologies developments (FUND, M36); **D3.3** Application of PHTS results (UPOL, M36).**; D3.4** Plant soil management strategies report (UNIBO, M36)**; D3.5** Report of sustainable solutions manufacturing for pilot plant trials (CHEM, M40).

Work Package No.	**WP4**	Starting date / Ending date	M12 / M48
Work Package title	**Pest and disease management integration. Leader: ANEC**		

Objectives

- Implement the technologies developed and carry out plant bioassays trials against CLso pest disease in an intended environments beyond Europe (e.g., Ecuador).
- Validate the postharvest technology and soil treatment solution at TRL5.
- Carry out a techno-economic analysis, environmental and social benefits.
- Define measures for the implementation of new policy initiatives and decisions.

Description of Work

T4.1 Validation field trials against CLso pest disease under real condition in Ecuador (M20-M48). Leader: INIAP. Participants: CHEM.

Subtask 4.1.1 Testing potato varieties against BC and CLso. *Leader: INIAP.*

CLso-infected BC collected from Zebra Chip affected research plots in INIAP Research Station (Sta. Catalina, Ecuador) and maintained at INIAPs facilities will be used for these experiments. Selected potato varieties (at least 30) from T 1.3 and other origins (Europe) will be tested under greenhouse and field conditions for resistance/tolerance to BC and CLso. Under greenhouse to inoculate plants, five potato psyllids from a INIAPs laboratory colony will be released at the base of individual plants within each cage (40 psyllids/cage). Choice and non-choice experiments for BC will be developed according to T1.4.2 procedure. The best materials for resistance/tolerance to BC and CLso will be selected for field test under natural conditions, information about severity, incidence, number of BC adults, nymphs will be recorded based on weekly monitoring and chip tuber quality will be evaluated.

Subtask 4.1.2 Validation of digital tool under natural conditions of CLso infections in potato. *Leader: INIAP.*

Digital tool build up for the early detection and surveillance of CLso infections in potato plants, will be tested under greenhouse conditions with confirmed potato CLso infected with BC at different phenological development stages, plants with different symptoms expression and tubers, compared with healthy controls. Under field conditions digital tool previously tested will be validated in potato plants and tubers infected with ZC.

Subtask T4.1.3 Validation of free natural solutions against BC. *Leader: INIAP.*

In *in vitro* potato plantlets will be planted into individual pots in greenhouse, BC adults obtained from an CLso-positive INIAPs colony will be reared on potato plants. At least six biorational insecticides solutions manufactured by CHEM with potential to control BC will be chosen to carry out bioassays on potato plants. Adult behaviour, mortality number of adults, eggs and nymphs will be measured on leaflets. The biorational products with better control of BC will be evaluated under field conditions using large plots (at least 2 ha).

ECSA supported by INIAP will hold two co-creative workshops with potato small-scale farmers in Ecuador. In the first one, an overall dissemination of the PataFEST project will be shared amongst the participants explaining the strategy approaches and solution to implement. In the second workshops, farmers will have the change to carry out free trials of a selected biorational insecticides solution on potato plants in their facilities and evaluate the repellent effectiveness against BC.

T4.2 Validation of postharvest technologies under real conditions in European potato warehouses (M20-M48). Leader: AB. Participants: FUND, CHEM, BOEH, ECSA, TERREA.

Subtask T4.2.1 Validation of postharvest biocontrol solution. *Leader: TERREA.*

Validation of postharvest technologies will be carried out in AB (UK), TERREA (France) and BOEH (Germany) warehouses on two ware potato varieties including organic crops: one common potato variety (Vivaldi or La Vie) for all the countries involved and a second one variety different selected for each geographical area.

CHEM will ship solutions to warehouses and application will be done by dipping potato in a tank or surface sprayed and let dried before the entrance in the storage chambers. At least 2 tons of potato (per year) will be tested and stored during conventional storage period. Trials will be validated on both organic and conventional potato tubers during two postharvest seasons.

Subtask T4.2.2 Validation of portable tools for postharvest warehouse. *Leader: BOEH.*

VOCs portable sensor devices developed in WP3 will be also validated by the same partners in potato warehouses. Potato samples will be placed in potato vented boxes and introduced into a storage chamber (100Kg potato). To concentrates organic volatiles generated from potato samples, pallets will be wrapped into semipermeable (avoiding the complete absence of oxygen) or introduced in a methacrylate storage chamber. Real time VOCs monitoring will be done during a defined period of 6-7 months per each potato postharvest season. Trials will be validated during two potato postharvest seasons. Periodically, presence of soil-borne pathogen incidence (presence or absence), severity (% of total surface affected) and quality determinations as sucrose and reducing sugars will be determined.

App digital tool developed in WP2 for the early detection of postharvest pathogen developed will be also validate in potato tubers stored in warehouse. During the storage, the app will be used to acquire information from the pathogen evolution, periodically. Validation will be done during two potato postharvest seasons.

T4.3 Evaluation of soil treatment solutions and potato plant pesticide free natural solutions under real cultivation conditions in European countries (M20-M48). Leader: UDAPA. Participants: CHEM, UNIBO, BOEH, ECSA, ANEC, TERREA, AB.

Validation of soil treatment solutions will be carried out in UDAPA (Spain), TERREA (France) and BOEH (Germany) under real conditions. Solutions will be shipped by CHEM evaluated on at least 2 ha including both organic and conventional potato crops during two growing seasons. Periodically, soil quality will be determined by UNIBO analysing macronutrients content (N, P, K) and presence of soil-borne pathogens (metagenomics analysis) for the evaluation of soil health.

Potato pesticide free natural solution will be also tested in European potato fields by UDAPA (Spain), TERREA (France) BOEH (Germany) and AB (UK) against aphids' threats. Pesticide free natural solution will be tested on plant by spraying application on 1 ha of fields. Natural solution effectiveness was measured by the presence of aphids (number of captures) through locating insect sticky traps close to the plant both treated and control.

Besides, ECSA and ANEC will hold **three co-creative workshops** with citizens that have urban gardens in Berlin-Bradenburg region and small horticulture crop farmers in Spain. During these workshops, PataFEST will be introduced and the opportunity to carry out free trails of natural solutions for plant and soil will be provided to the

participants. By this way, citizens will feel involved in the PataFEST scientific activities and during workshop their concerns and questions related the integration of strategies to overcome emerging crops pest and disease threats will be addressed to PataFEST expert partners.

T4.4 Technical and economic assessment (TEA), Environmental Life Cycle Assessment (LCA) and Social LCA (M12-M48). Leader: AINIA, Participant: All.

Subtask T4.4.1 Technical and economic assessment. *Leader: AINIA.*

Technical viability & readiness for each strategy will be individually validated with the support of the partners involved in the project. AINIA will analyse the technical and economic feasibility of the three different strategies developed in previous WPs. Outcomes will be assessed via a sensibility and risk analysis to determine potential hot spots limiting the feasibility of a technology and elucidating the best strategy for industrial implementation. Cost estimations and profitable analysis will be conducted.

Subtask T4.4.2 Environmental Life Cycle Assessment, Social Life Cycle Assessment and Life Cycle Cost Analysis. *Leader: AINIA.* Participant. All

This subtask embodies the assessment of the potential environmental impacts of the three strategies tested in PataFEST according to ISO 14040/14044. AINIA will develop specific questionnaires to collect inventory data from experimental applied strategies developed in previous WPs. The environmental impact categories to be included in the analysis are among others: GHG emissions, primary energy demand, and resource depletion, and complemented with additional impact categories based on EU ILCD and PEF methods. The social impacts and socio-economic benefits of PataFEST strategies will be assessed following the UNEP-SETAC guideline for S-LCA. The activity variables will be determined, subcategories for the LCSA will be defined and key stakeholders will be identified. The S-LCA will follow a similar structure and will be conducted in parallel to the E-LCA. The total cost of ownership including the capital, operational costs and cost reductions resulting for the PataFEST strategies will be analysed by performing an LCC analysis in parallel to the LCA (in terms of functional unit and approach). The analysis will be developed in two steps: cost identification and cost benefits analysis. Based on obtained CAPEX and OPEX, financial indicators (e.g., NPV) will be calculated to assess the economic feasibility of PataFEST.

T4.5 Policy Recommendations (M12-M48). Leader: EUPAT. Participant: All.

The aim of this task is to provide feedback to policy measures generated by the project that will contribute to designing, monitoring, reviewing, and rectifying (if necessary) existing policy and programmatic measures or shaping and supporting the implementation of new policy initiatives and decisions. For this purpose, the main current EU policies will be identified, and their different aspects of sustainable development will be analysed and assessed, including how to incorporate social and economic aspects into those policies. To create real impact with their work, all partners will translate the policy recommendations into their national language(s) and will widely disseminate them among relevant stakeholders in the policy and decision-making cycle. Most of them ANEC, EUPAT, JRC, etc. have several years of experience in lobbying and advocacy work, they can make use for this of already existing networks and contacts. The same applies for the European recommendations. These recommendations will be sent to the EC, the European Parliament, the Committee of the Regions and multiplicator European networks, the Commission Expert Group on Plant Health Legislation.

Deliverables: D4.1 Validation field trials against CLso pest disease under real condition in Ecuador (INIAP, M48); **D4.2** Validation of postharvest technologies tested in European potato warehouses (AB, M48); **D4.3** Validation of natural and sustainable solutions for potato plant and soil under real cultivation conditions (UDAPA, M48); **D4.4** Report on TEA, LCA and LCC methodologies (AINIA, M30, M48); **D.4.5** Policy recommendation report (EUPAT, M40).

Work Package No.	**WP5**	Starting date / Ending date	M1 / M48
Work Package title	**Communication, Dissemination and Exploitation (Leader: KNEIA)**		

Objectives

- Enable potential future exploitation of the project results by disseminating them to the relevant stakeholders.
- Ensure project's findings are widely communicated to the scientific community and the general public.
- Document undertaken and proposed dissemination and communication activities.
- Ensure the project results reach the relevant stakeholders who will use and implement them.
- Guarantee IPR management according to the Guidelines of the EC within the Consortium Agreement
- Ensure clustering and coordination with other related EU/National projects and initiatives.
- Attract business partners and potential stakeholders and to create market demand.

Description of Work

T5.1 Communication activities (M1-M48). Leader: EUPAT. Participants: All.

Subtask 5.1.1 Communication Plan and reporting on communication: *Leader: EUPAT.* EUPAT, with the support of KNEIA and FUND, will draw up a Communication Plan, as part of the PDEC (D5.3), to mobilise stakeholders'

interests and secure their support to the project. This plan will be closely linked with the dissemination activities, and it will follow the steps described in section 2.2.

Subtask 5.1.2 Project website: *Leader: KNEIA.* A project's website will be developed by KNEIA and managed by KNEIA, EUPAT and FUND. It will serve as a communication tool to all target audiences and it will host a restricted access area for partners to store information and facilitate file sharing and internal communication. The public site will present the project brand and non-confidential project information, including infographics presenting the strategic, competitive and economic benefits, the PataFEST open access publications, etc. The platform will enable to disseminate non-confidential results (public reports, deliverables, publications as part of our green open access, etc.). It will also house on-line training tools and educational materials and will incorporate 2 official project videos. At the end of the project, the relevant public information and results will be kept as the central virtual hub for post-project activities.

Subtask 5.1.3 Communication and promotional materials: *Leader: EUPAT.* Project leaflets, posters, project video, newsletters and promotional materials (gadgets for tradeshows, etc.) will be designed by KNEIA with the strong support of EUPAT and shared with all partners in order to support project communication and branding efforts. **Two dissemination videos –** an introductory video on M6 for the general public and a dissemination report video at the end of the project describing the final results- will be developed. They will be published on PataFEST's main communication channels (YouTube, website, and social media profiles) as well on EUPAT and other project's partners' channels to raise awareness and engage stakeholders with project pilots and to promote uptake after the end of the project. These materials will be reported and Annexed to the Communication Plan as part of the PDEC.

Subtask 5.1.4 Social/digital media channels: *Leader: KNEIA.* This subtask will exploit the potential of reaching our targeted audiences through social media channels: a social media grid will be drawn up; it will plot the most appropriate social media channel for targeting specific audiences. Appropriate pages will be designed for each channel and a content maintenance plan will be put in place for managing the streams of information to secure and maintain followers. Social and digital media will contribute to create 'communities of support' to the project. Tools such as Twitter, LinkedIn and YouTube will be used. Visual media, videos, animations, icons and info-graphic imagery, and importantly, mobile enabled content and richer content experiences will be developed by KNEIA with the support of EUPAT and other partners.

T5.2 PataFEST dissemination (M1-M48). Leader: KNEIA. Participants: All.

A dissemination plan will be developed as part of the PDEC, building on the preliminary plan in Section 2.2 and in complete alignment with the principles of **Open Science**. The following activities will be executed: **i)** Publishing at least **6/8 scientific articles** in journals and conferences (after preliminary IP protection check); **ii)** generation of at least **60 "practice abstracts"** to contribute to EIP-AGRI repository; **iii)** organising special webinars and scientific workshops; **iv)** targeted placement of articles and features in relevant trade and industry focused publications; **v)** organisation of the "PataFEST Agenda" of events that will be planned for the 48-month project duration: trade fairs, exhibitions, workshops, seminars, conferences, round tables, etc., that the consortium is already planning to attend or organising as part of ongoing activities. The target is **24 existing events** as well as the organisation of dedicated project workshops and webinars: **2 intermediate workshops** with brother projects, a **final conference** (M48) will be organised in Spain (FUND and KNEIA) to showcase results/impacts, **3 citizen science workshops** to test PataFEST's natural solutions for plant and soil in citizens' own urban gardens and farmers (T4.3) and **8 citizen science webinars** (ECSA).

T5.3 Exploitation of results and Business model (M1-M48). Leader: KNEIA. Participants: All.

A draft of the Exploitation Plan will be delivered as part of the PDEC (D5.3,). Built on the information already provided in Section 2.2, the **Exploitation Plan** will be structured as follows: **i)** the planned key exploitable results and their expected key applications: development status, facts and figures that facilitate the assessment of the potential impact, differences from existing competing products/services, main IPR issues to be clarified in the consortium agreement; **ii)** the exploitation team: partners involved in the exploitation activities providing information on their previous experience and business interests, complementing external resources if not in the consortium; **iii)** the market: the potential geographical and economic size of the market, market trends, main competitors and competitive advantage; **iv)** Business model including marketing strategy and main clients: will be elaborated using CANVAS methodology adapted to the relevant target markets. Results ownership list will be included in the final periodic report as well as in the final version of PDEC.

T5.4 PataFEST Clustering (M1-M48). Leader: KNEIA. Participants: All.

This task will ensure that PATAFEST is networked with relevant national and international projects, initiatives and networks. Networking with other projects will ensure knowledge interchange and reuse between previous and future projects. A close interaction with other proposals selected for funding under the same or closer topics will be ensured. The networking plan will take into account relevant projects within international stakeholders and other EU/national projects (some of them identified in Section 1.2). Most of the project partners are members of relevant networks, groups and associations. These memberships will be instrumental to promote cooperation and integration with PataFEST. Members of consortia of the networked projects will be invited to the 2 intermediate workshops (Task 5.2) and exchange of (non-sensitive) information will be implemented (e.g., by exploiting the newsletter or through dedicated meetings). One section of the Newsletter (Task 5.1) will be dedicated to the information on other ongoing

projects and exchange of newsletters from the other connected projects will be pursued. The progress of this task will be reported in the PDEC.

T5.5 Multi-Actor Participatory Platform (MAPP) (M1-M48). Leader: KNEIA. Participants: All.

The project will develop the MAPP (D5.4), an ad hoc online platform within PataFEST's website to ensure identification, engagement, inclusion and active participation of relevant stakeholders of the potato value chain and their diverse perspectives (including the **Quadruple Helix** –Government, Industry, Academia and Civil Society- plus policy makers, relevant associations and coordinators of other projects in the same topic). The MAPP will act as instrument for knowledge sharing and engagement with both other successful operating and future HE projects. The role of this platform will be to organise, structure and offer a place for knowledge sharing for supporting users in their valorisation decisions, while collecting feedback and expectations of relevant stakeholders & end-users on the project activities and results. Thus, a dedicated section with all the citizen of science activities will dedicated. This collected information will contribute to define common development campaigns aimed to unlock the potentials of the project exploitation strategies in different EU Countries. All stakeholders identified will be invited to participate in an online survey designed to measure stakeholder characteristics, knowledge, interests and expectations relevant for the project. Based on the outcome of the survey, more targeted dissemination and exploitation actions will be implemented and relationships built with key stakeholders to the project. Survey results will feed the revision of the PDEC (D5.3).

Deliverables: D5.1 Project Website (KNEIA, M6); **D5.2:** Project Videos (KNEIA, M6, M48); **D5.3** PDEC - Plan for the dissemination, exploitation and communication activities (KNEIA, M6, M24, M36); **D5.4:** Multi-Actor Participatory Platform (KNEIA, M30, M48).

Work Package No.	**WP6**	Starting date / Ending date	M1 / M48
Work Package title	**Project Management (Leader: FUND)**		

Objectives

- Coordinate and manage the project and the efficient, smooth, and timely execution of all activities.
- Ensure an accurate and on time communication flow among the project partners and with the EC to track the project progress and meet its objectives.
- Complete the project within the agreed time schedule and budget.

Description of Work

T6.1 Administrative & Financial Management (M1-M48). Leader: FUND; Participants: All.

FUND will be responsible for the administrative coordination of the project, delivery and follow-up of all administrative documents required by the EC, and keeping the consortium up to date of any information communicated by the EC. All the administrative procedures will be defined both in the consortium agreement and on the project manual; this will cover management, reporting, documents, review, general quality assurance, standards, etc. The quality control will be important to check that all contributions and conclusions are consistent and meet the requirements for the deliverables, reports, and other documents. The coordinator will be responsible for the economic coordination of the project, including follow-up and delivery of all financial documents required by the EC and will distribute the EC contribution amongst the partners according to the GA and CA.

T6.2 Scientific, decision-making and risk management (M1-M48). Leader: NEIK; Participants: All.

The main decisions of the project will be taken by the Scientific Coordinator (NEIK). All decisions will be implemented through regular 6-monthly consortium meetings. The project progress and any deviations will be reported to the project coordinator continuously, and to the Steering Committee by half-yearly meetings, to evaluate the Project's scientific progress and any need of eventual corrective measures. The risks and mitigation/contingency measures in Section 3.1.4 will be continuously monitored and updated.

T6.3. Data management plan and data interoperability (M1-M48). Leader: FUND; Participants: All.

A Data Management Plan (DMP) will be developed to ensure that scientific research data will be easily discoverable, accessible, assessable, and intelligible, useable beyond the original purpose for which they are collected, as well as interoperable to specific quality standards. Procedures for handling research data, data to be shared and published in open access, how data will be stored and preserved (during and after the project) and security measures to be implemented to prevent unauthorised access to project data and documents will be defined.

Deliverables: **D6.1** Project manual: Guide for the execution of the technical and administrative work (FUND, M6); **D6.2** Data Management Plan and update version (FUND, M6, M24, M48).

3.1.1 List of work packages

WP	Work Package Title	No	Leader	P/M	Start Month	End month
1	Pest monitoring and resistance potato varieties selection	2	NEIK	118	1	48
2	Early detection and surveillance tools	4	IRIS	223	1	48
3	Prevention strategies based on sustainable solutions	1	FUND	190	1	40
4	Pest and disease management integration strategies	15	ANEC	175	12	48

5	Dissemination, communication and exploitation of results	5	KNEIA	126	1	48
6	Project Management	1	FUND	50	1	48
			Total	**882**		

3.1.2 List of deliverables

Del. #	Deliverable name	WP	Participant	Type	Diss. level	Delivery date
D1.1	Monitoring results of CLso spreading pattern	1	UPOL	R	CO	M36
D1.2	Catalogue of Potato RGs with SNP information	1	NEIK	R	CO	M18
D1.3	Genotypic Potato Chip data and SPET data	1	NEIK	R	CO	M30
D1.4	Association study results, predicted performance data and selection of most promising RG for pest and postharvest diseases	1	NEIK	R	CO	M36
D1.5	Result databases and bioinformatics tool for publication and dissemination actions.	1	NEIK	OTHER	PU	M48
D2.1	Identification of phenotyping traits in infected plants and tubers.	2	UPOL	R	CO	M36
D2.2	VOCs biomarkers identification and quantification results	2	CRUN	R	CO	M36
D2.3	Development of a portable VOCs sensor for potato warehouses	2	FUND	R	CO	M36
D2.4	Alpha version of PATAFEST digital platform design development for validation	2	IRIS	OTHER	PU	M20
D2.5	DSS design, development, and optimisation for the early detection of potatoes diseases and pest spreading	2	IRIS	OTHER	PU	M42
D2.6	AI-based algorithms developments for the real-time detection of potato diseases and prediction of pest and disease spreading	2	FUND	OTHER	CO	M48
D3.1	Natural plant solution with repellent and anti-adhesive properties developments	3	TECHN	R	CO	M36
D3.2	Postharvest biocontrol coating solutions and controlled atmosphere technologies developments	3	FUND	R	CO	M36
D3.3	Application of PHTS results	3	UPOL	R	CO	M36
D3.4	Plant soil management strategies report	3	UNIBO	R	CO	M36
D3.5	Report of sustainable solutions manufacturing for pilot plant	3	CHEM	DEM	CO	M40
D4.1	Validation field trials against CLso pest disease under real condition in Ecuador	4	INIAP	DEM	CO	M48
D4.2	Validation of postharvest technologies tested in European potato warehouses	4	AB	DEM	CO	M48
D4.3	Validation of natural and sustainable solutions for potato plant and soil under real cultivation conditions	4	UDAPA	DEM	CO	M48
D4.4	Report on TEA, LCA and LCC methodologies	4	AINIA	R	CO	M30, M48
D4.5	Policy recommendation report	4	EUPAT	DEC	PU	M40
D5.1	Project Website	5	KNEIA	DEC	PU	M6
D5.2	Project Videos	5	KNEIA	DEC	PU	M6, M48
D5.3	PDEC - Plan for the dissemination, exploitation and communication activities	5	KNEIA	R	CO	M6, M24, M36
D5.4	Multi-Actor Participatory Platform	5	KNEIA	DEC	PU	M30, M48
D6.1	Project manual: Guide for the execution of the technical and administrative work	6	FUND	R	CO	M6
D6.2	Data Management Plan and update version	6	FUND	DMP	CO	M6, M24, M48

R: Report; DEC: Dissemination, exploitation, and communication; DMP: Data management plan; DEM: Demonstrator, pilot, prototype, plan designs; OTHER: Software, technical diagram, algorithms, models, etc..; CO: Confidential; PU: public.

3.1.3 List of milestones

M#	Description	WP(s)	Date	Means of verification
M1.1	Analysis of CLso spread pathway	WP1	24	At least 80% of CLso samples will be successfully sequenced.
M1.2	Catalogue of mapped RGs with integrated SNP	WP1	12	At least 80% of the RG target resources will be exploited.
M1.3	Polymorphic Genotypic Potato Chip Data and SPET data	WP1	26	Molecular characterisation of 350 germplasm collection (SNP chip array and SPET) will be analysed.

M#	Description	WP(s)	Date	Means of verification
M1.4	Bioassays tolerances characterisation and validation against CLso, psyllid vector and postharvest disease.	WP1	30	100 potato samples will be characterised against the three target agents.
M1.5	Associate analysis and compilation and integration resource using bioinformatics platform.	WP1	38	80% of target tolerances resources databases and trait association studies will be integrated in the platform.
M2.1	Identification and monitoring of phenotyping traits (biomarkers) through PHTS as a technology to diagnose the potato health plant.	WP2	26	At least 15 different morphological and/or physiological as biomarkers will be successfully identified using HTPS technology.
M2.2	Development of high-performance portable VOCs sensor and digital tool field application for the real time early detection of ZC and potato pathogen disease.	WP2	36	Sensors will be able to detect at least 2 identified VOCs biomarkers. At least 80% reliability detection performances by using mobile application.
M2.3	Fitting and prediction performance of AI algorithms for the early detection of plant pest and postharvest pathogens; AI algorithm of pest and disease spreading based on climate change and globalisation.	WP2, WP4	30	Different means of verification to obtain information for different case of use and decisions to evaluate. F1: 0.85-0.9; ROC-AUC (receiver operating characteristic-area under curve): 0.8-0.85; PR-AUC (precision recall-area under curve): 0.75-0.8
M3.1	Effectiveness of plant natural and sustainable solution with CLso repellent action; biocontrol solution and controlled atmosphere strategies for the postharvest pathogen reduction.	WP3, WP4	30	Application of pesticide free natural solutions will reduce the presence of psyllid (BC) by at least 50% and the incidence of ZC by 25%. Combination of biocontrol coatings and controlled atmosphere storage treatments will decrease at least 40% the incidence of soil-borne pathogen.
M3.2	Effective soil management strategy to increase plant health status.	WP3, WP4	30	Soil management strategies will reduce incidence of CLso and post-harvest pathogens in potato crop by 30%.
M3.3	Sustainable solutions manufacturing at pilot plant.	WP3	30	Pesticide free solution for potato plant; biocontrol coating and alternative biofertilizer for soil prototypes will be manufactured producing at least 100L per solution.
M4.1	Testing technologies in plant bioassays trials against CLso pest disease in an intended environments beyond Europe (Ecuador).	WP4	40	At least 20 potato varieties having resistant traits against CLso will be evaluated in field trials; 2 ha of potato plants will be treated using pesticide free natural solutions against CLso vector; digital tool will be in field performing 90% reliability on the diagnostic detection of CLso disease in potato crops.
M4.2	Validation of the postharvest technology and soil treatment solution at TRL5.	WP4	40	VOCs portable sensors will be validated on potato (100 Kg) under real postharvest warehouse storage condition; postharvest biocontrol solution will be tested on 1 tons of potato; digital tool will be operated and validated under real conditions.
M4.3	Integrated TEA, LCA and LCC assessment .	WP4	36	At least 80% of input and data received from all WPs will delivered by partners

M#	Description	WP(s)	Date	Means of verification
M5.1	Development of online multi-Actor Participatory Platform	WP5	36	Engagement with other successful projects under the addressed HE call and other future HE projects.
M5.2	Deploying dissemination activities	WP5	36	At least 45 Practise abstract published; a total of 15 (national and international) events organised, and 5 citizen science activities.

3.1.4 Critical risk for implementation

Description of risk (indicate level of (i) likelihood, and (ii) severity: Low/Medium/High)	WP(s)	Proposed risk-mitigation measures
Low resolution of the 16S gene to infer spread patterns (i) Low, (ii) Low.	WP1	Deeper sequencing to stable backbone of the phylogenetic tree.
Failure in obtaining at least 300 Potato RGs with integrated SNP. (i) Low, (ii) Medium.	WP1	Analysis of additional, recent Potato Resistance gene /QTL studies as well as SNP studies which are coming up during the project duration.
VOCs concentration and detection method does not differentiate between control and infected tubers. (i) Low, (ii) Medium.	WP2	General VOCs markers with potentially less resolution will be used as alternative to differentiate between infected and control tubers (e.g., acetone)
No identification and monitoring of phenotyping traits (biomarkers). i) Low, (ii) High.	WP2	Five different sensors (RGB, FluorCam, IR, two types of hyperspectral) will be used, each able to analyse from 3 to 20 different morphological and/or physiological traits. Multivariate statistical approaches will be used to identify the most informative traits.
High interference detected by VOCs sensor. i) Medium, (ii) Medium.	WP2	E-nose technology would be developed in order to improve the selectivity and accuracy of the array to make predictive measures
Inability to generate user adoption for the mobile app. (i) Medium, (ii) High.	WP2	To facilitate adoption, seamless user experience will be ensured, deliver quick value to the user and also employ gamification techniques for onboarding users to the platform.
Inability to access data from Third Party Apps. (i) Low, (ii) High.	WP2	Project coordinator will approach the owners/managers of these Third-Party applications to propose an integration (the ability for the PataFEST mobile app to be a catalyst for an open ecosystem that promotes data sharing would be a valuable and impactful measure).
Poor performance of algorithm fit modelling (< 0.7 AUC). (i) Medium, (ii) High.	WP2	Accuracy readjustment of data, in order to increase, correctly classified points by a total number of points on the test set. Being: Accuracy = #correctly classified points / #misclassified points.
Low effectiveness of plant, soil, and biocontrol postharvest solutions. (i) Low, (ii) Medium.	WP3	Higher concentrations of bioactive ingredients will be tested and/or alternative functional compounds will be included in the enhanced formulation prototypes.
Extreme weather conditions leading to high temperatures in the glasshouse. (i) Medium, (ii) Medium.	WP3	Avoid running during summer months; use vertical screens in the glasshouse to reduce effect.
Introduction of other diseases affecting crop development and yield. (i) Medium, (ii) Medium.	WP3	An agreed crop management plan will be in place prior to the start of the experiment; close daily monitoring of crop for early detection and prompt response.
Low performance properties of the sustainable solutions manufactured at pilot plant. (i) Medium, (ii) High.	WP3	Manufacturing processing steps will be reviewed and adjusted at pilot plant level productions.
No presence of ZC infected potato plants under field and greenhouse conditions. (i) Low, (ii) Low.	WP4	Maintain an infected CLso and BC colony under INIAP´s facilities

Description of risk (indicate level of (i) likelihood, and (ii) severity: Low/Medium/High)	WP(s)	Proposed risk-mitigation measures
Potato varieties selected not suitable for the trial validations. (i) Low, (ii) Low.	WP4	Other potato varieties will be selected according to their availability and higher susceptibility to be infected by pathogens.
Data or information delay for the sustainability and techno-economic assessment. (i) Medium, (ii) Medium.	WP4	The progress will be monitored throughout the project and delays will be identified early and addressed. Further, strong internal communication will ensure that partners are clear on when and what information is expected delivered.

3.1.5 Summary of staff effort

Participant		WP1	WP2	WP3	WP4	WP5	WP6	TOTAL
1	**FUND**	2	85	**37**	10	10	**20**	164
2	**NEIKER**	**40**	4	3	3	3	8	61
3	INIAP	14	0	0	18	3	1	36
4	**IRIS**	0	**72**	0	1	3	1	77
5	**KNEIA**	0	0	0	1	**48**	1	50
6	CRUN	0	21	30	1	5	6	63
7	UDAPA	0	0	0	15	3	1	19
8	CHEMIA	0	0	15	1	3	1	20
9	UPOL	52	40	35	1	3	1	132
10	TECHN	0	0	33	1	3	1	38
11	EUPAT	0	0	0	5	5	2	12
12	UNIBO	0	0	35	1	3	1	40
13	BOEH	0	0	0	12	3	1	16
14	ECSA	5	1	1	10	12	1	30
15	**ANEC**	5	0	0	**20**	10	1	36
16	TERREA	0	0	0	12	3	1	16
17	AB	0	0	0	12	3	1	16
18	AINIA	0	0	0	51	3	1	55
		118	**223**	**190**	**175**	**126**	**50**	**882**

3.1.6 Subcontracting cost

3. INIAP → 14,000 €: Sequencing service for CLso detection; BC colony maintenance (service).
7. UDAPA → 2,000 €: Soil analysis and plant virus analysis.
17. AB → 7,250 €: Albert Bartlett grows potatoes on subcontracted land and sub-contracts long term commercial storage.
18.AINIA → 50,000 €: Social-LCA analysis subcontracting.

3.1.7 Purchase cost items (travel and subsistence, equipment, and other goods, works and services)

2.NEIKER	**Cost, € 112,000**	**Justification** (only partners which the sum of the costs for 'travel', 'equipment', and 'other goods and services' exceeds 15% of the personnel costs)
Travel	30,000	2 Researchers attendance to 2 annual meetings/year (Nº of travels= 16); 4 Researchers attendance to 1 National Congress / year (Nº of travels= 16); 2 Researchers attendance to 1 International Congress (EAPR, ALAP)/ year (Nº of travels= 8)
Other goods & Services	82,000	WP1: SNP GGP V4 chip hybridations, SPET Analysis, DNA extractions, Materials for multiplication and bioassays; WP5: Materials for Publication, Dissemination, Open access fees, Congresses Inscription Fees; WP6: Host of 1 Project Meeting and CFS cost.
3.INIAP	**69,100**	
Travel	12,000	Travels to some project meetings and technical meetings for validation activities in selected locations in Ecuador
Equipment	7,100	INIAP´s equipment). Partial time dedicated to project activities: Molecular lab equipment and weather data logger, 25% and 20% allocated to the project, respectively.

Other goods & Services	50,000	Consumables and supplies for the activities of evaluation of germplasm and validation of genotypes and products for B.cockerelli control in controlled and field conditions.
4.IRIS	**100,990**	
Travel	11,200	7 Project meetings, installations and dissemination events attendance during the project.
Equipment	15,850	Depreciation of computing prototypes: laptops (x2), android or IOS smartphones and/or tablets (x2), for use over the project lifetime for the development, hosting, testing and validation of the digital tools and apps; bug fixing. Sensorics and related equipment including wiring.
Other goods & Services	73,940	Servers and Cloud hosting fees and internal services over the course of the project. Database and interface management, licensing and security, adaptability and connectivity related costs with readers. Tutorials & platform optimisation & updates. Market analysis and exploitation analysis related costs (5 K). Host of 1 project meeting. CFS cost (6 K).
5.KNEIA	**69,200**	
Travel	13,200	8 project meetings x 2 people. 2 dissemination travel per 2 people
Other goods & Services	56,000	Project website (stock images, domain, other developmen costs9; Dissemination materials (poster +leaflets+ pen drives); Videos (production expenditures); Conference organisation (Presential and Hybrid conferences); Conference attendance fees; Market Surveillance.
7.UDAPA	**26,500**	
Travel	10,500	8 project meetings x 1 person; National travels to assist to field trials, participation in 1-2 commercial fairs x 2 persons.
Other goods & Services	16,000	Costs to cover the field trials: Potato seed, irrigation, fertilizers, pesticides, cultivation costs, harvest and storage; Visitor fees to commercial fairs + 2 persons.
8.CHEMIA	**25,000**	
Travel	9,000	8 project meetings x 1 person; National travels to assist to field trials, participation in commercial fairs: 2 international and 2 national (Italy).
Other goods & Services	16,000	Material to prepare the formulations: solvents, surfactants, active ingredients, small laboratory materials (10 k€); Shipping costs of the formulations (100 L --> 400 € to Europe; to Ecuador 1000: 5 k€ in total for all the shipment).
9.UPOL	**203,935**	
Travel	50,135	3 technical meetings within WP1, WP2 and WP3 (1 meeting/person/WP); 10 travels to participate in scientific conferences to present results of WP1, WP2 and WP3; 24 travels to 2 project meeting/year, including Kick off meeting for 3 representatives of WP1, WP2 and WP3.
Other goods & Services	153.800	1) expenses covering consumables and material for PHTS in WP1, WP2 and WP3, needed for technical optimisations and upgrades of the phenotyping devices for PHTS (electronic materials, HW components, RGB cameras) and the phenotyping experiments in OloPhen platform (multiwell plates and special pots for cultivation of the potato plants, chemicals and growth media for in vitro propagation), 2) services needed for maintenance and upgrades of the OloPhen platform for the capacity allocated to the project, 3) Library preparation kit consumables and consumable laboratory chemicals, buffers, plastics, and chemicals related to qPCR for WP1; 4) WGS sequencing services for WP1.
10.TECHN[57]	**93,500**	
Travel	19,000	2 Technical trips to Ecuador; 4 trips to fairs (1 per year); 4 Trips for Dissemination (1 per year). Project meetings.
Other goods & Services	74,500	Chemicals and reagents; media for microbiological assays, fluorescent probes; disposable laboratory consumables. CFS Cost.
12.UNIBO	**55,500**	
Travel	10,500	International and national symposiums, 8 persons; 6 project meetings.

[57] For partner Technion, such Costs include special purchase of services to be purchased from its fully owned and controlled entity Technion Research & Development Foundation Ltd. ("TRDF"), under the provisions of Article 6.2.C.3 of the Grant Agreement. Additionally, the costs mentioned under "Personnel Cost" and "Summary of Staff Effort", also include costs of personnel seconded by TRDF to Technion (the Beneficiary), under the provisions of Article 6.2.A.2 and A.3 of the Grant Agreement. All costs above constitute an integral part of the total Technion budget in the Project. All services, items and secondments will be provided and put by TRDF at the disposal of Technion, at the premises of Technion, so Technion carries carry out the work of the Project.

Other goods & Services	45,000	Raw material (agricultural waste, soil, pots, fertilizers, plants), Lab consumables (reagents, disposable plastics) + soil microbiota analysis; Fees to Open Access to peer reviewed scientific publications; Host of 1 Project Meeting and CFS cost.
16.TERREA	**18,000**	
Travel	10,500	Participation in commercial fairs (potato europe, world potato congress, fruit logistica); 8 consortium meeting/year, including Kick off meeting x 1 person.
Other goods & Services	7,500	Potato use in the warehouse (~ 5 tons), tools for field trials, quality analysis (sugars), fee visitor to commercial fairs.
17.AB	**25,572.5**	
Travel	13,312.5	National travels to the field and storage facilities for trials, Participation in 2 commercial fairs x 2 persons; 8 travels/person (2 consortium meeting/year, including Kick off meeting) x 1,5 persons.
Other goods & Services	12,260	Potato seeds, irrigation, Cultivation, Harvest, Fertilizer, Pesticides (e.g.blight control) (assumption 1 ha: 60Tons harvested potato used for trials and not onward sale); Visitor Fee to participate in 2 commercial fairs x 2 persons.

3.1.8 Internally invoiced good and services

3. INIAP → 10,000 €: Multiplication of plant selected materials (in vitro & tubers) for characterisation of resistance/tolerance, Lab analysis for Clso diagnosis; Multiplication of selected genotypes for validation experiments.
6. CRUN → 91,744.30€: CHAPs rental for facilities and Agri-EPI Rental to grow potato tubers under controlled conditions simulating different climate scenarios.
10. TECHN → 187,450 €: Funds are requested for Technion infrastructure equipment services dedicated to materials characterisation as well as facilities of microbiology infrastructure. These include facilities for imaging such as SEM, TEM, FIB, confocal microscopy, etc. Facilities for chemical and structural characterisation (such as TGA,XRD, etc.); facilities for surface and topography characterisation (such as AFM, XPS etc.). Microbiological infrastructure includes usage of autoclaves, greenhouse facilities, plate-readers, etc. As an example, an hour of TEM service costs 400 NIS, SEM and XRD costs 300 NIS per hour. As characterisation is a major part of the research, we are requesting about 58,000 euro per year. This is equivalent of ~650 hours per year on average of characterisation and microbiology facilities usage (for 45 weeks per year, this is under 15 hours per week).
12.UNIBO → 4,000 €: Use of 1 box at the greenhouse centre (Bologna University) for experimental plots trials. Use of testing facilities for chemical and phytosanitary analyses.

3.2 Capacity of participants and consortium as a whole

The full R&D potential of PataFEST consortium is met in the preliminary phases of the proposal preparation by identifying the main barriers, challenges and problems that the actors working across the potato sector, farmers, consumers and citizens are currently facing. In order to build up the project's objectives and maximise our results and outcomes, Funditec, as coordinator, has involved technology suppliers, potato producers, agricultural/potato associations and cooperatives, and citizens/consumers at the early phases of the design of the work packages. This multi-actor approach, along with the partners' complementing skills will speed up the creation and adoption of the Integrated Pest Management strategy from the start of the project and over time. In addition, the Multi-Actor Participatory Platform will also be essential for promoting rural community engagement at all scales (local, national, and worldwide) and for spreading the project's proposed measures against emerging potato pests after project's end. This is possible because PataFEST strategically involved 2 agricultural cooperatives (UDAPA and ANECOPP) and the EUROPATAT association working very closely with the farmers, traders, and professionals' experts of potato and other vegetables production. PataFEST consortium includes **18 partners** from **9 different countries** (6 from EU and UK, Israel and the international collaboration established with INIAP from Ecuador). PataFEST has a **strong representation of SMEs, 33%** of the consortium partners (IRIS, KNEIA, UDAPA, CHEM, BOEH, TERREA) and **large companies** (**11%,** ANEC, AB) which is accompanied by **4 RTOs** (**22%:** FUND, NEIK, INIAP and AINIA), **4 Universities (22%:** CRUN, UPOL, TECHN and UNIBO) and **2 associations** (**17%:** EUPAT and ECSA). The capacity of each partners is critical for the implementation of this project:

Scientific/Technical suppliers

RTO ↓ **Coordinator** **Spain** ↓ *Postharvest biocontrol solutions* *AI algorithm prediction model* *Portable volatiles sensor*	FUNDITEC is a RTO that develops innovate projects in its own laboratory facilities located in Madrid (currently running 9 EU projects, one as coordinator). FUNDITEC has three strategic research areas Advanced Materials, AgriFood and Digital Transformation, all of them participating in PataFEST. The leader of the project is **Dr. Christian Guidelli**, Principal Researcher of the **Agrifood Department and coordinator of the project.** In addition, he will be responsible of the postharvest biocontrol solutions development. He has more than 10 years of experience on the development of edible coatings with antioxidant and antimicrobial properties for meat, fresh and minimally processed fruits and vegetables and in the development of alternative and natural methodologies to extend the food shelf life. Two other researchers participate in the project: **Dr. Alejandro Echeverria** and **Dr. Jose Ignacio Lozano**. Alejandro is a Computer Engineer and PhD in Computer Engineering and Telecommunications (specialising in artificial intelligence and evolutionary computation) and will be in charge of the development of **AI algorithms for prediction models**. Jose Ignacio is PhD in organic electrochemistry and expert in managing of technological innovation. He has a vast knowledge in developing applied research for industrial processes at large scale, development of smart materials, catalyst and industrial processing technology. He is in charge of the development of the **portable volatile sensors**.
RTO ↓ **Scientific coordinator** **Spain** ↓ *RGs identification* *Molecular Breeding*	**NEIKER** is specialised in creating innovative solutions for the agro-livestock and forestry sectors. The scientific team of NEIKER belongs to the Department of Plant Health and Production (**Dr. Amaia Ortiz, Dr. Enrique Ritter, Dr. Jose Ignacio Ruiz de Galarreta** and **Dr. Leire Barandalla**) and is composed by researchers with a long trajectory in the developing of new potato varieties adapted to different agro-climatic conditions and resistance to viruses, fungi and bacteria. NEIKER has an annual budget of 13 m€ and performs an average of 120 R&D projects. In order to complete all foreseen tasks in PATAFEST, an extended Bioinformatics platform including Software is available for data analyses, over 200 m^2 of fully automated greenhouses for performing bio-assays and over 70 Ha land for field trials.
RTO **Ecuador** ↓ *Characterisation of germplasm Field and bioassays*	**INIAP** is the National Agricultural Research Institute of **Ecuador**. INIAP's main research lines are plant breeding, agro-biotechnology, integrated crop and livestock management and conservation and use of genetic resources. INIAP has 7 experimental stations and 5 experimental farm throughout Ecuadorian territory. In this project, the validation of plant bioassays against CLso pest disease will be located in one of these experimental farms. **Dr. Xavier Cuesta** and **Dr. Carmen Castillo** are the INIAP's researchers involved in this project, both are experts in potato plant breeding and the psyllid affecting potato crops in South America (> 100 scientific publications). Their deep knowledge and expertise of potato's threats that are already affected in this country is crucial to achieve the objectives of PataFEST.
University **Israel** ↓ *Vector repellent and antifungal solutions*	**Technion- Israel Institute of Technology** is Israel's first university and its largest center of applied research. Technión is ranked as the top university in Israel, listed among the top 100 universities in the world, and is also famous for its 3 Nobel prizes in chemistry. **Prof. Boaz Pokroy** of the **Faculty of Materials Science and Engineering** (expert biomineralisation and bio-inspired surface engineering) in and *Prof. Ester Segal* of the **Faculty of Biotechnology and Food Engineering** (expert in nanomaterials their interface microorganisms) will lead the Technion's activity in PataFEST. They both demonstrate outstanding research record (>120 scientific publications each) and vast experience in EU projects (including ERC starting grant, ERC PoCs participation, and coordination of several EU projects). Their labs house state-of-the-art infrastructure for the design, fabrication and detailed characterisation of biomimetic protective plant coatings, including equipment for contact angle measurements, electron and confocal microscopes, well equipped microbiology laboratories, and greenhouses.
RTO **Spain** ↓ *LCA & LCC*	**AINIA** is a private non-profit agrifood technological centre with one of the largest business social bases in Europe which is supported by its participation in the following EU-granted projects: LIFE ECODHYBAT (awarded as the Best of the Best Environment Project in 2018), LIFE LEMNA, SUSCLEAN (FP7-KBBE) URBIOFIN and CHEERS, amongst many other projects. AINIA have a wealth of experience in the application of LCSA methodologies to assess environmental and economic impacts of new biorefinery models, bioproducts and related technologies by using updated versions of the highly-recognised software for LCSA purposes such as GaBi software, GaBi Professional database, Ecoinvent (specialised in E-LCA) and some GaBi extended databases. In PataFEST, **Dr Alfredo Rodrigo** will be the researcher in charge of the LCSA within the project. He has more than 23 years of experience as project manager on environment issues related to food industry.

University UK ↓ *Volatile biomarkers identification Storage technology Plant soil management*	Three groups of **Cranfield Soil, Agrifood and Biosciences Centre (CSAFI)** participate in PataFEST: **Plant Science Laboratory** (**Dr. Carmen Alamar,** lecturer in postharvest biology), **Soil Systems Group** (Dr. **Wilfred Otten**, professor of soil biophysics) and **Applied Mycology Group** (Prof. **Angel Medina**, the head of the Applied Mycology Group and the Director of Environment and Agrifood Theme). This centre has an internationally recognised profile for its research and teaching in Postharvest Technology, Applied Mycology, Soil Science, and Agri-informatics and it hosts the environmental analytical Facilities which will provide access to state-of-the-art facilities for full physiological and metabolomics studies of both non-volatile and volatile organic compounds, as well as molecular biology laboratories. Cranfield University houses four CHAP environmental simulation rooms (growth rooms) and a glasshouse that can be used to grow potato tubers under controlled conditions (viz., temperature, light, and CO_2) simulating different climate scenarios. The glasshouse hosts a Lemnatec phenotyping gantry for high resolution plant phenotyping. Additionally, soil lysimeters (1 m^3 and 0.5 m^3) can be used within the glasshouse for pilot scale studies where control field conditions can be simulated (e.g., drought and flood) and monitored throughout the experiment.
University Czech Republic ↓ *Phenotyping study for portable sensor development*	**Palacky University Olomouc (UPOL)** is the second oldest university in the Czech Republic (founded 1573) educating around 25000 students. In PataFEST, the research will be performed in the **Czech Advanced Technology and Research Institute - Center of the Region Haná (CATRIN – CRH),** focused on plant research and the development of plant biotechnologies. PataFEST research will be performed in two different research groups: **Biodiversity and Molecular Evolution** (**Prof. Ladislav Bocák** and **Dr. Michal Motyka**) and in **Phenotyping** (**Dr. Lukás Spíchal** and **Dr. Nuria de Diego**). CATRIN-CRH, as a leading member of the Czech Plant Phenotyping Network (CzPPN) runs OloPhen, a plant phenotyping platform for automated high-throughput screening equipped with different sensors for non-invasive analysis of morphological and physiological traits.
University Italy ↓ *Biobased solutions for soil management*	From the University of Bologna, the **Department of Agricultural and Food Sciences (Ass. Prof. Claudio Ratti - Plant Health)** and the **Department of Chemistry (Prof. Giuseppe Falini**) participate. UNIBO has an active patent portfolio of over 370 patents related to more than 140 inventions. It also has an active portfolio of more than 20 patented plant varieties, 80% of which are increased in value through active licenses with geographic coverage in countries in different parts of the world. Thanks to its scientific excellence and a central project management support working closely, UNIBO has gained a considerable experience in International and European research projects UNIBO is partner of the EIT Knowledge & Innovation Communities and many of the most important EU initiatives related to the Horizon 2020 framework program (i.e. ETPs, EIPs, JPIs, etc.). At EU level, UNIBO is part of the Scientific Advisory boards of JPI HDHL and FACCE JPI, member of the EIP AGRI Focus Group Water & Agriculture, ETP Ocean and Biofuels and RTO member of the BIC Consortium supporting the JTI Biobased Industry.
SME Spain ↓ *Modelling and data processing Image analysis digital tool developer*	The experience of **IRIS** (>45 M€, >100 projects, >15 patents, ranked as the 6th SME in Europe and 1st in Spain in participation in H2020) as an Advanced Engineering company specialised in **Real-Time Monitoring solutions** is crucial to carry out the image analysis for early detection and surveillance. IRIS' technologies are based on Photonics-based Process Analytical Technology systems and Industrial Artificial Intelligence (Machine Learning and Deep Learning) for data-driven process efficiencies and quality standardisation in the Food, Chemical, Pharmaceutical, Bio-based and other process industries. IRIS team of 70 multidisciplinary experts combines experienced programmers, engineers (optical, mechatronic, electronic and process engineers), chemists, physicists, mathematicians and data scientists, biologists and technologists, converging the scientific and technological fields required for Industry 4.0 solutions development.
SME Italy ↓ *Plant and postharvest solutions*	**CHEMIA's** core business is the production of pesticides and, leading the Italian manufacture of products of all toxicological classes and the formulation of solutions with different properties. Thanks to the wide range of formulates, Chemia is the perfect partner in PataFEST to formulate the new products following the highest standard of quality, safety and respect for the environment.

Agricultural & Potato Associations and Cooperatives

Spain & Belgium ↓	**UDAPA**, **ANECOOP** and **EUROPATAT** are key players in this project as they bring relevant actors that will be benefit of the outcomes of the project. With these three partners, the project will reach directly to more than 300.000 professionals of potato and other vegetables. **UDAPA**, is a cooperative in Basque Country (Spain) aiming to professionalize the production processes and sale of fresh potatoes. UDAPA has a 14.000 m^2 where the project will validate the soil

Validator of plant soil management strategies *Multi-actor participatory platform* *Dissemination & Communication*	treatment solutions and potato plant pesticide free natural solutions. **ANECOOP** is Spain's largest agricultural production network of vegetables and fruits with 73 member cooperatives and companies across the country (€790.3 million turnover and 845.410 tonnes of production in 2020-2021). **ANECOOP** has a strong commmitment to innovation participating in EU projects, including Operational Groups (Gocitrus and Climate Change in Stone Fruits) with a strong focus on adaptation to climate change, detection of emerging pests (RFF-IPM) and used of agricultural residues to for high added value applications (Pro-enrich) and 20 Ha for experimental field trials and Biotechnology laboratories. **EUROPATAT** has 70 years representing EU potato traders (seed, ware and early potato). Currently, it has **62 members** that represent 20 countries in the EU and beyond. All of them will ensure the profesional interest of the sector, facilitate the collection of data and information towards all the members, influence policy decision-makers about the project's outcomes and promote networking and communication activitities.

Potato Producers

Germany, UK, France ↓ *Validators of plant soil management and postharvest strategies*	The project includes three potato companies from the main EU potato producers as the validators of the soil and postharvest strategies directly in their facilities: **Kartoffel-Böhmer** (Germany, over 30 Ha greenhouse and 250 Ha outdoor production) is a reliable company for food retailers, farmers and producers for many decades both at national and international potato markets. Böhmer has outdoor and indoor organic production of potatoes which is crucial for the trials of PataFEST solutions. **TERRÉA** (France) is a SME that belongs to VITALIS network. VITALIS is made up of 6 companies which altogether are leading the French trading and packaging of potatoes, onions and condiments. **Albert Bartlett** is the UK's leading supplier of potatoes. Albert supplies potatoes from their wide network of 62 farmers and growers across UK.

Citizens and stakeholder engagement, communication, dissemination & exploitation

SME,Spain. *Dissemination, Communication Exploitation*	**KNEIA** is a SME specialised in providing support to research organisations along the research value chain with a special focus on environmental and circular economy approaches. Kneia is expert in scientific dissemination, communication, open science and market research in EU projects. Kneia has currently 4 Horizon projects and 8 H2020 projects.
Association Germany ↓ *Citizen of Science*	The European Citizen Science Association (**ECSA**) promotes citizen science as a tool for evidence-based policy and sustainable development. They support a strong and growing citizen science community in Europe and beyond thanks to its wide science network with different background. ECSA's role in PataFEST is crucial to deploy the citizen science approach requested in the topic to ensure that the results are also citizen-generated. ECSA will create relevant training activities to engage citizen at all levels. ECSA is algo the legacy organisation of the EU-Citizen.Science Platform, which provides resources, tools, training, community forums and information on projects, where PataFEST will joint.